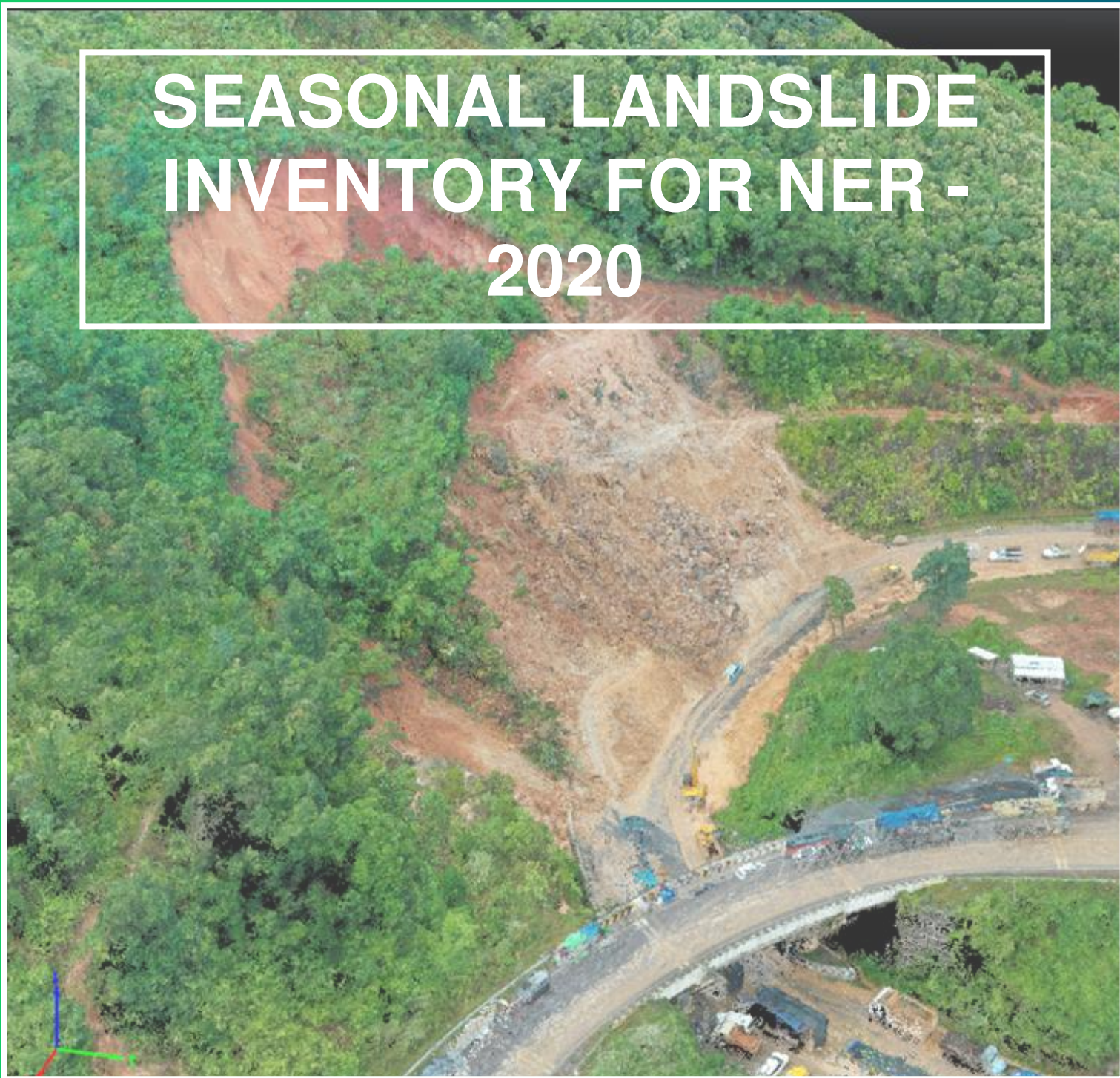


SEASONAL LANDSLIDE INVENTORY FOR NER - 2020



North Eastern Regional Node for Disaster Risk Reduction (NERDRR)

NORTH EASTERN SPACE APPLICATIONS CENTRE

GOVERNMENT OF INDIA, DEPARTMENT OF SPACE,

UMIAM-793103, MEGHALAYA

December, 2020

2020

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Government of India

For more information: Visit the NERDRR website - <https://www.nerdr.gov.in/>

Document Control Sheet

1	Security Classification	Unclassified		
2	Distribution	Open		
3	Type of Document	Technical Report		
4	Document Control Number	NESAC-SR-261-2021		
5	Title	Seasonal Landslide Inventory for the North Eastern Indian Region (NER) – 2020		
6	Number of Pages	64		
7	Author(s)	NERDRR Team for Landslide, NESAC		
8	Scrutiny Mechanism	Compiled by: NERDRR Team for Landslide NESAC	Reviewed/Approved by: Dr. KK Sarma, (DPD, NERDRR)	Controlled by: Director, NESAC
9	Sponsor(s)	North East Space Applications Centre/Department of Space		
10	Date of Initiation	May, 2020		
11	Date of Publication	December, 2020		
12	Abstract	The study was taken up as an ongoing process under NERDRR programme of NESAC, Umiam. It is basically a compendium of all reported landslide incidents collected from various sources, namely, Newspapers, Online Media, SDMA Reports, GSI Reports and Field Visits. A total 141 landslide incidents have been documented for the different states of the NER for the year 2020, majority of which have been triggered by rainfall. Further, a total of 52 casualties have been reported out of which 12 are from Arunachal Pradesh, 25 are from Assam, 7 are each from Meghalaya and Sikkim and 1 is from Nagaland. In addition, collapse of road-sections of National/State Highways are reported from Manipur, Meghalaya, Nagaland and Sikkim. This report also contains information on 24 hrs total rainfall prior to each event. Some of these locations are verified in terms of precise latitude and longitude, either through field survey, post-dated imagery or through an available report from GSI or SDMA. Accordingly, all codification has been assigned. The systematic inventory thus compiled will help in carrying out future landslide research. Further this will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency of landslide and magnitude of landslide incidents in an area and the NER as a whole. More information and data is available in the NERDRR portal (https://www.nerdr.gov.in)		
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PREFACE

North Eastern Regional node for Disaster Risk Reduction (NER-DRR) has been set up at NESAC, Umiam in the year 2011 to provide scientific solutions towards disaster management to the entire region. Almost entire NER consisting of 8 states, namely, Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura is susceptible to landslides except the plain areas of Assam, Manipur and Tripura. Physiographically NER is a part of Extra-Peninsula of India and of geologically recent origin, having exposed rocks from Archaean to Quaternary ages. This diverse setup makes the terrain susceptible to landslides. While, intense and/or prolonged monsoonal rainfall and seismic activities are the immediate trigger, anthropogenic factors have a role in accelerating the issue. Hence, the need for immediate and systematic cataloguing of landslide incidents is realized by creating a geospatial database highlighting the landslide locations with respect to nearby settlements and transportation network after the event is reported.

It is expected that, the systematic inventory thus compiled will help in carrying out future landslide research. Further, visualization of the same in spatial context will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency and magnitude of landslide incidents in an area and NER as a whole.

DPD, NERDRR

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1. INTRODUCTION

The term "landslide" describes a wide variety of processes that result in the downward and outward movement of slope forming materials composed of natural rock, soils, artificial fills or combinations of these material, (Verns, 1958). According to Geological Survey of India (GSI) reports, 427 k sq. km area accounting 12.6% of the Indian landmass is landslide prone and this contributes app. 8% of global landslide fatalities. Hence, the people living in the North Eastern Region of India are very familiar with both the magnitude and the frequency with which landslides wreak havoc each year. NASA's Landslide Hazard Assessment for Situational Awareness Model (LHASA) classifies Landslide Susceptibility in large parts of North East India under categories 4 and 5 (in a scale from 0 to 5).

Under the purview of North Eastern Regional Node for Disaster Risk Reduction (NERDRR), a need for immediate cataloguing of landslide incidents and creation of a Geospatial database has been realized by highlighting the landslide locations with respect to nearby settlements and transportation network after report of any landslide incident. The systematic inventory thus compiled will help in carrying out future landslide research. Further this will help in assessing State/area wise risk in terms of yearly loss of life and property, disruption of transportation network, frequency of landslide and magnitude of landslide incidents for the whole of NER. This brief report provides information on reported landslide incidences pan NER collected from various sources. However, this may not be inclusive of all the events that have triggered during 2020 monsoon season.

2. SOURCES OF DATA

2.1 LANDSLIDE DATA

Unlike other disasters like earthquakes, so far, landslide incidents are not instrumentally recorded and reported anywhere immediately. Consequently, immediate cataloguing of landslide incidents has been done mainly from newspapers and online/social media. Subsequently, landslide locations are verified in terms of precise latitude and longitude either through field survey, post-dated satellite/UAV imagery or through published report from GSI and SDMA as and when available. Accordingly, all codification is assigned. Thus, all the sources can be grouped into four categories:

1. Newspapers and Media
2. SDMA Reports
3. GSI Reports
4. Field Visit

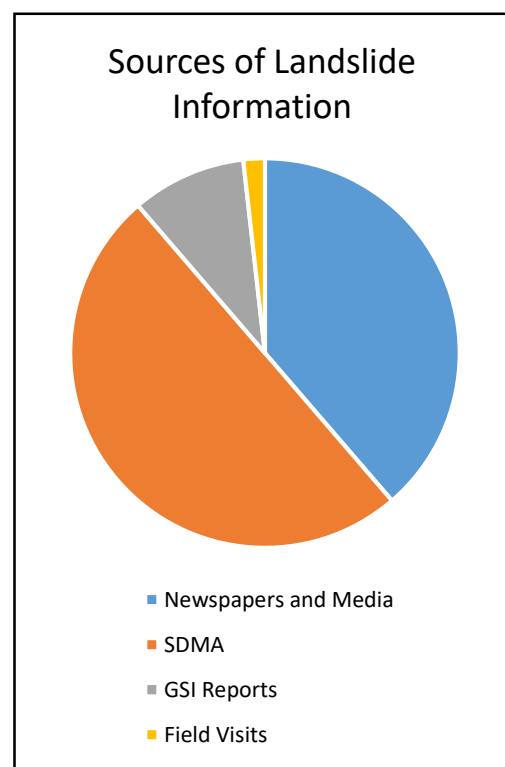


Figure 1: Sources of landslide incidents

Due care is taken to minimise the confusion regarding date of occurrences and precise incident locations of reported events.

Source	Assam	Arunachal Pradesh	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim
Total Landslide Events	20	5	15	23	41	10	27
Newspapers & Media	02	3	12	13		10	25
SDMA	18				41		25
GSI Reports		2	3	9			2
Field Visits				1*			

Table 1: Sources of landslide data for different states of NER

* Due to Covid-19 situation, in 2020 field visits are not taken up

2.1.1 NEWSPAPERS AND MEDIA:

Newspapers are the primary sources of information that record landslide incidents in the NER. Hence, leading daily newspapers distributed in the different states of NER were thoroughly reviewed either through hard-copies or in the online archives. (Annexure II).

These days, online media plays a significant role in reporting disaster incidents. Therefore, data collected from newspapers was supplemented with information acquired from online media.

State wise list of newspapers that are referred for 2020 landslide incidents in NER are provided in Table 2.

Newspaper and Media	Assam	Arunachal Pradesh	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim
The Shillong Times	✓	✓		✓			
The Assam Tribune	✓						
The Sangai Express			✓				
The Sikkim Express							✓
Eastern Mirror Nagaland						✓	
The Morung Express						✓	
Youtube Clips		✓	✓	✓		✓	

Table 2: State wise list of newspapers reporting landslide incidents in NER

2.1.2 SDMA:

State Disaster Management Authorities (SDMA) headed by the Chief Ministers of respective states also keep a record of disastrous incidents in their respective states. Database related to landslide are regularly being shared by 8 SDMA's of NER to NESAC, Umiam under the ongoing National project, '**Landslide Early Warning Systems (LEWS)**' being executed by NRSC, Hyderabad. Information on disastrous events is also available on SDMA portals. SDMA data are also used for final validation of positional accuracy of the landslide locations as well as date and time of occurrence (Annexure III).

2.1.3 GSI REPORTS:

The Geological Survey of India is the nodal agency for landslide related studies in India. On their web portal, under Recent Incidences 2020, details of event-specific investigations undertaken by GSI are also available post field surveys. GSI reports are referred mostly for final validation of positional accuracy of the landslide locations as well as date and time of occurrence (Annexure IV).

2.1.4 FIELD:

A field visit was carried out to landslide site that has occurred near Thangshalai Village, Ri Bhoi district, Meghalaya on 27th September 2020. This incident led to the closure of the Shillong Bypass for more than 2 weeks. Data from this field visit was also added in the inventory. However, due to Covid-19 restrictions field visits are not carried out to Other NER states.



Figure 2: UAV image of Thangshalai Landslide site, Meghalaya

2.2 SATELLITE DATA

The Satellite imagery available post event or post monsoon are used to extract landslide areas through polygon features to validate positional accuracy of the landslide locations. Freely available imagery from SENTINEL 2 Multispectral Instrument (MSI) is used extraction and updation of landslide features. Details of the Level 1C and the Level 2A products of SENTINEL 2 MSI used for this study is given below.

Source: ESA Website

The [Level-1C](#) product is composed of 100x100 km² tiles (ortho-images in UTM/WGS84 projection). The Level-1C product results from using a [Digital Elevation Model \(DEM\)](#) to project the image in cartographic geometry. Per-pixel radiometric measurements are provided in Top Of Atmosphere (TOA) reflectances along with the parameters to transform them into radiances. Level-1C products are resampled with a constant Ground Sampling Distance (GSD) of 10, 20 and 60 m depending on the native resolution of the different spectral bands. In Level-1C products, pixel coordinates refer to the upper left corner of the pixel.

Level-1C products will additionally include [Cloud Masks](#) and [ECMWF](#) data (total column of ozone, total column of water vapour and mean sea level pressure).

The [Level-2A](#) product provides Bottom Of Atmosphere (BOA) reflectance images derived from the associated Level-1C products. Each Level-2A product is composed of 100x100 km² tiles in cartographic geometry (UTM/WGS84 projection).

Level-2A products are systematically generated at the ground segment over Europe since March 2018, and the production was extended to global in December 2018.

Figure 3: Sentinel 1C and 2A

Both Level 1C and 2A products were used with the Ground Sampling Distance (GSD) of 10m.

2.3 RAINFALL DATA

Rainfall data from NASA's Global Precipitation Mission (GPM) has been used, considering the seamless availability of rainfall data for the entire NER. 'IMERG LATE RUN' product of spatial resolution - 0.1 degrees/~10km has been used.



GLOBAL PRECIPITATION MEASUREMENT

- Missions
- Data
- Applications
- Science
- Resources
- Education

IMERG Late Run

- Near real-time gridded global multi-satellite precipitation estimates with quasi-Lagrangian time interpolation

Overview:

This algorithm is intended to intercalibrate, merge, and interpolate "all" satellite microwave precipitation estimates, together with microwave-calibrated infrared (IR) satellite estimates, precipitation gauge analyses, and potentially other precipitation estimators at fine time and space scales for the TRMM and GPM eras over the entire globe. The system is run several times for each observation time, first giving a quick estimate (IMERG Early Run) and successively providing better estimates as more data arrive (IMERG Late Run). The final step uses monthly gauge data to create research-level products (IMERG Final Run).

The main difference between the IMERG Early and Late Run is that Early only has forward propagation (which basically amounts to extrapolation), while the Late has both forward and backward propagation (allowing interpolation). The additional 10 hours of latency allows lagging data transmissions to make it into the Late run, even if they were not available for the Early.

Figure 4: GPM IMERG Late Run Product

3. CHARACTERISTICS OF THE LANDSLIDE INVENTORY

The inventory is seasonal in nature and contains events triggered by events of 2020 monsoonal rainfall.

A total of 141 landslide incidents have been documented in the different states of NER. 52 casualties are reported out of which 12 from Arunachal Pradesh, 25 from Assam, 7 each from Meghalaya and Sikkim and 1 from Nagaland.

In addition, collapses of road-sections along National/State Highways have been reported from Meghalaya, Manipur, Nagaland and Sikkim.

Base and other ancillary information were used to correlate the inventory database is of 1:50,000 scale.

This database reflects the landslide locations immediately after a landslide event is reported in newspaper, online/offline media, preferably within 24hrs of reporting, so depending upon the availability of precise location information the inventory database is grouped into following 2 categories: -

- **TYPE 1 (Represented by Green colour)**

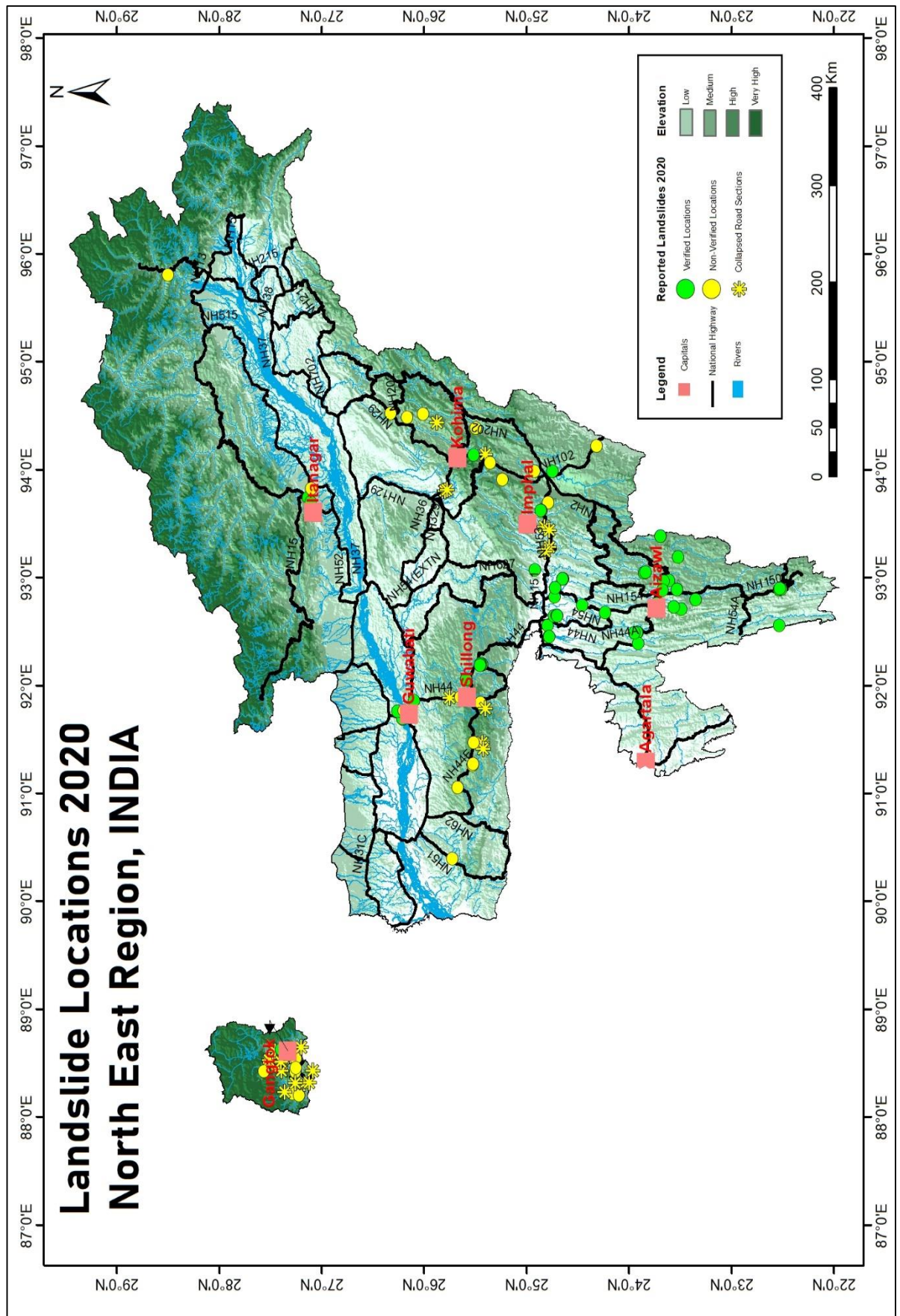
These are events for which precise latitude/longitude information is available. Such events are generally validated from SDMA/GSI reports and/or Field Visits. Post-dated satellite imageries are also used for validation purpose where spatial resolution of the image and size and exposure condition of the landslide makes it possible. (E.g. Passingdong Landslide in Sikkim).

- **TYPE 2 (Represented by Yellow colour)**

These are events for which precise latitude/longitude information not available rather is referred with respect to nearby village/settlement or locality name by newspapers/media reports. In most of the case these slides are of smaller dimension and could not be visualized using freely available Sentinel 2 data of 10m spatial resolution. However, as and when needed the same may be validated by using post-dated high resolution satellite data.

- A 3rd category (Type 3) includes areas where effect of landslide on road sections along National Highways/State Highways is reported, which has affected transportation. Separate symbology is used to distinguish these locations. However, challenge associated for validation of these locations is that, road sections usually get restored or repaired at the earliest possible time thus only a small window is left for validation purpose either through a field visit or using satellite imagery. Revisit status of particular satellite further complicates the situation. E.g. – Khongnem Thana Landslide. However, as and when needed the same may be validated by using post-dated satellite data of appropriate resolution.

All the landslide locations also have standard attribute information, in terms of unique ID, location, date/time/type of events, casualty information, 24hrs rainfall prior to the event etc. (Annexure V). Due care is being taken to validate and convert all the event locations to Type 1 at the earliest.



Map 1: Landslide Incidents in the NER 2020

Table							
2020_LS_points							
Slide ID	Source	District	Lat	Long	Date of Ev	Casualty	Status
Newspaper&Media/2020/Anurachal Pradesh/25	Newspaper&Media	Papum Pare	27.104	93.826	17-Sep-20		
Newspaper&Media/2020/Manipur/26	Newspaper&Media	Tamenglong	24.774	93.284	2-Jun-20		
Newspaper&Media/2020/Manipur/27	Newspaper&Media	Tamenglong	24.8205	93.473	2-Jun-20		
Newspaper&Media/2020/Manipur/28	Newspaper&Media	Tamenglong	24.8172	93.2457			
Newspaper&Media/2020/Manipur/29	Newspaper&Media	Senapati	25.396	94.142	2-Jun-20		
Newspaper&Media/2020/Manipur/30	Newspaper&Media	Senapati	25.236	93.911	6-Jul-20		
Newspaper&Media/2020/Manipur/31	Newspaper&Media	Imphal east	24.748	93.986	11-Aug-20		Verified from Satellite Imagery
Newspaper&Media/2020/Manipur/32	Newspaper&Media	Senapati	24.925	93.985	21-Jun-20		
Newspaper&Media/2020/Manipur/33	Newspaper&Media	Senapati	25.498	94.387	20-Aug-20		
Newspaper&Media/2020/Manipur/34	Newspaper&Media	Senapati	24.792	93.696	28-Aug-20		
Newspaper&Media/2020/Manipur/35	Newspaper&Media	Chandel	24.318	94.222	20-Aug-20		
Newspaper&Media/2020/Manipur/36	Newspaper&Media	Tamenglong	24.775278	93.438889	27-May-20		
Newspaper&Media/2020/Manipur/37	Newspaper&Media	Senapati	25.355956	94.065278	23-Oct-20		Verified from Satellite Imagery
GSI_REPORTS/2020/Manipur/38	GSI_REPORTS	Tamenglong	24.855361	93.624028	Last week of June		Verified by GSI
GSI_REPORTS/2020/Manipur/39	GSI_REPORTS	Tamenglong	24.862694	93.624056	Last week of June		Verified by GSI
GSI_REPORTS/2020/Manipur/40	GSI_REPORTS	Senapati	25.515917	94.136528	Sometime before 21 July		Verified by GSI
GSI_REPORTS/2020/Meghalaya/41	GSI_REPORTS	East Khasi Hills	25.62	91.882722	21-May-20		Verified by GSI
GSI_REPORTS/2020/Meghalaya/42	GSI_REPORTS	Jaintia Hills	25.442333	92.194472	26-May-20		Verified by GSI
GSI_REPORTS/2020/Meghalaya/43	GSI_REPORTS	Jaintia Hills	25.441778	92.195583	26-May-20		Verified by GSI
GSI_REPORTS/2020/Meghalaya/44	GSI_REPORTS	Jaintia Hills	25.441389	92.191083	26-May-20		Verified by GSI
GSI_REPORTS/2020/Meghalaya/45	GSI_REPORTS	Jaintia Hills	25.436306	92.19525	26-May-20		Verified by GSI
GSI_REPORTS/2020/Meghalaya/46	GSI_REPORTS	Jaintia Hills	25.435056	92.198361	26-May-20		Verified by GSI
GSI_REPORTS/2020/Meghalaya/47	GSI_REPORTS	Jaintia Hills	25.447417	92.201389	26-May-20		Verified by GSI
GSI_REPORTS/2020/Meghalaya/48	GSI_REPORTS	Jaintia Hills	25.45075	92.194528	26-May-20		Verified by GSI
GSI_REPORTS/2020/Meghalaya/49	GSI_REPORTS	Jaintia Hills	25.453517	92.187847	26-May-20		Verified by GSI
Newspaper&Media/2020/Meghalaya/50	Newspaper&Media	East Khasi Hills	25.448	91.84	11-Jul-20		
FIELD_VISIT/2020/Meghalaya/51	FIELD_VISIT	East Khasi Hills	25.590667	92.059889	27-Sep-20		Verified by Field Visit
Newspaper&Media/2020/Meghalaya/52	Newspaper&Media	Ri-bhoi	25.746	91.889	24-Oct-20		
Newspaper&Media/2020/Meghalaya/53	Newspaper&Media	West Khasi Hills	25.43	91.471	10-Jul-20		
Newspaper&Media/2020/Meghalaya/54	Newspaper&Media	West Khasi Hills	25.417	91.416	9-Jul-20		
Newspaper&Media/2020/Meghalaya/55	Newspaper&Media	West Khasi Hills	25.528	91.262	11-Jul-20		
Newspaper&Media/2020/Meghalaya/56	Newspaper&Media	East Khasi Hills	25.581	91.884	12-Jul-20	2 dead	
Newspaper&Media/2020/Meghalaya/57	Newspaper&Media	East Khasi Hills	25.561	91.88	25-Sep-20	5 dead	Verified from local personnel
Newspaper&Media/2020/Meghalaya/58	Newspaper&Media	West Garo Hills	25.726	90.394	24-Jul-20		
Newspaper&Media/2020/Meghalaya/59	Newspaper&Media	West Khasi Hills	25.523	91.272	14-Jul-20		
Newspaper&Media/2020/Meghalaya/60	Newspaper&Media	West Khasi Hills	25.516	91.469	14-Jul-20		
Newspaper&Media/2020/Meghalaya/61	Newspaper&Media	West Khasi Hills	25.676	91.055	14-Jul-20		
Newspaper&Media/2020/Meghalaya/62	Newspaper&Media	East Khasi Hills	25.65	91.893	24-Sep-20		
Newspaper&Media/2020/Meghalaya/63	Newspaper&Media	East Khasi Hills	25.4	91.786	29-May-20		
SDMA/2020/Mizoram/64	SDMA	Aizawl	23.731928	92.760028	06.10.2020		Verified by SDMA
SDMA/2020/Mizoram/65	SDMA	Aizawl	23.721556	92.731186	22.09.2020		Verified by SDMA
SDMA/2020/Mizoram/66	SDMA	Aizawl	23.740453	92.721578	01.10.2020		Verified by SDMA
SDMA/2020/Mizoram/67	SDMA	Aizawl	23.720156	92.698094	05.06.2020		Verified by SDMA
SDMA/2020/Mizoram/68	SDMA	Aizawl	23.723239	92.700333	27.09.2020		Verified by SDMA
SDMA/2020/Mizoram/69	SDMA	Aizawl	23.751072	92.695947	13.08.2020		Verified by SDMA
SDMA/2020/Mizoram/71	SDMA	Aizawl	23.739519	92.71415	02.10.2020		Verified by SDMA
SDMA/2020/Mizoram/72	SDMA	Aizawl	23.720222	92.728675	04.07.2020		Verified by SDMA
SDMA/2020/Mizoram/73	SDMA	Aizawl	23.488881	92.708261	31.08.2020		Verified by SDMA
SDMA/2020/Mizoram/74	SDMA	Aizawl	23.720278	92.716983	10.06.2020		Verified by SDMA
SDMA/2020/Mizoram/75	SDMA	Aizawl	23.562675	92.726283	11.07.2020		Verified by SDMA
SDMA/2020/Mizoram/76	SDMA	Aizawl	23.730278	92.757661	02.02.2020		Verified by SDMA
SDMA/2020/Mizoram/77	SDMA	Aizawl	23.746222	92.723003	15.06.2020		Verified by SDMA
SDMA/2020/Mizoram/78	SDMA	Aizawl	23.739747	92.708728	17.07.2020		Verified by SDMA
SDMA/2020/Mizoram/79	SDMA	Aizawl	23.612047	92.974247	28.05.2020		Verified by SDMA
SDMA/2020/Mizoram/80	SDMA	Aizawl	23.350964	92.797306	10.07.2020		Verified by SDMA
SDMA/2020/Mizoram/81	SDMA	Champhai	23.692778	93.382778	08.08.2020		Verified by SDMA
SDMA/2020/Mizoram/82	SDMA	Champhai	23.692778	93.382778	01.08.2020		Verified by SDMA
SDMA/2020/Mizoram/83	SDMA	Lawngtlai	22.536667	92.887222	07.07.2020		Verified by SDMA
SDMA/2020/Mizoram/84	SDMA	Lawngtlai	22.536111	92.889722	03.08.2020		Verified by SDMA
SDMA/2020/Mizoram/85	SDMA	Serchhip	23.533611	92.891944	04.08.2020		Verified by SDMA
SDMA/2020/Mizoram/86	SDMA	Lawngtlai	22.533611	92.5575	03.08.2020		Verified by SDMA
SDMA/2020/Mizoram/87	SDMA	Lawngtlai	22.534167	92.889444	20.08.2020		Verified by SDMA
SDMA/2020/Mizoram/88	SDMA	Serchhip	23.533611	92.890833	24.08.2020		Verified by SDMA
SDMA/2020/Mizoram/89	SDMA	Lawngtlai	22.515556	92.896389	02.08.2020		Verified by SDMA
SDMA/2020/Mizoram/90	SDMA	Lawngtlai	22.527222	92.898889	03.08.2020		Verified by SDMA
SDMA/2020/Mizoram/91	SDMA	Aizawl	23.6625	92.9675	02.10.2020		Verified by SDMA
SDMA/2020/Mizoram/92	SDMA	Aizawl	23.85	93.0475	02.10.2020 □ 03.10.2020		Verified by SDMA
SDMA/2020/Mizoram/93	SDMA	Aizawl	23.855617	93.0475	17.08.2020		Verified by SDMA
SDMA/2020/Mizoram/94	SDMA	Aizawl	23.841111	93.051944	13.10.2020		Verified by SDMA
SDMA/2020/Mizoram/95	SDMA	Dhalai	23.838611	92.053611	08.10.2020		Verified by SDMA
SDMA/2020/Mizoram/96	SDMA	Aizawl	23.673333	92.885556	10.07.2020		Verified by SDMA
SDMA/2020/Mizoram/97	SDMA	Mamit	23.92	92.491111	10.06.2020		Verified by SDMA
SDMA/2020/Mizoram/98	SDMA	Mamit	23.916944	92.492222	11.07.2020		Verified by SDMA
SDMA/2020/Mizoram/99	SDMA	Mamit	23.924167	92.490278	11.06.2020		Verified by SDMA
SDMA/2020/Mizoram/100	SDMA	Mamit	23.912778	92.390556	11.06.2020		Verified by SDMA
SDMA/2020/Mizoram/101	SDMA	Mamit	23.948611	92.49	29.05.2020		Verified by SDMA
SDMA/2020/Mizoram/102	SDMA	Champhai	23.523083	93.193194	03.10.2020		Verified by SDMA
SDMA/2020/Mizoram/103	SDMA	Kolasib	24.231944	92.675278	21.10.2020		Verified by SDMA
SDMA/2020/Mizoram/104	SDMA	Kolasib	24.455	92.741667	25.05.2020		

Figure 5: Landslide Attributes in the Inventory

4. METHODOLOGY

4.1 LANDSLIDE POINT AND POLYGON EXTRACTION

For preparation of Seasonal Landslide Inventory post monsoon for the year 2020, a thorough review of articles relevant to landslide incidents from all the major Daily Newspapers in NER was carried out. The location of all landslide incidents was derived from newspapers and an initial database of landslide locations was prepared in the form of a Point Shapefile.

This initial database of landslide locations was checked against SDMA and GSI reports for verification and increasing the accuracy of the geolocation of landslide points.

Of all the landslide incidents, all those events which were determined large enough to be visualized on Sentinel 2 imagery were extracted and a 10km buffer was created around them (for ambiguity resolution in geolocation).

Finally using Pre and Post Event Satellite Imagery, landslide polygons within the 10km buffer wherever visible, were digitized.

The custom code used in the methodology has been attached as Annexure VI.

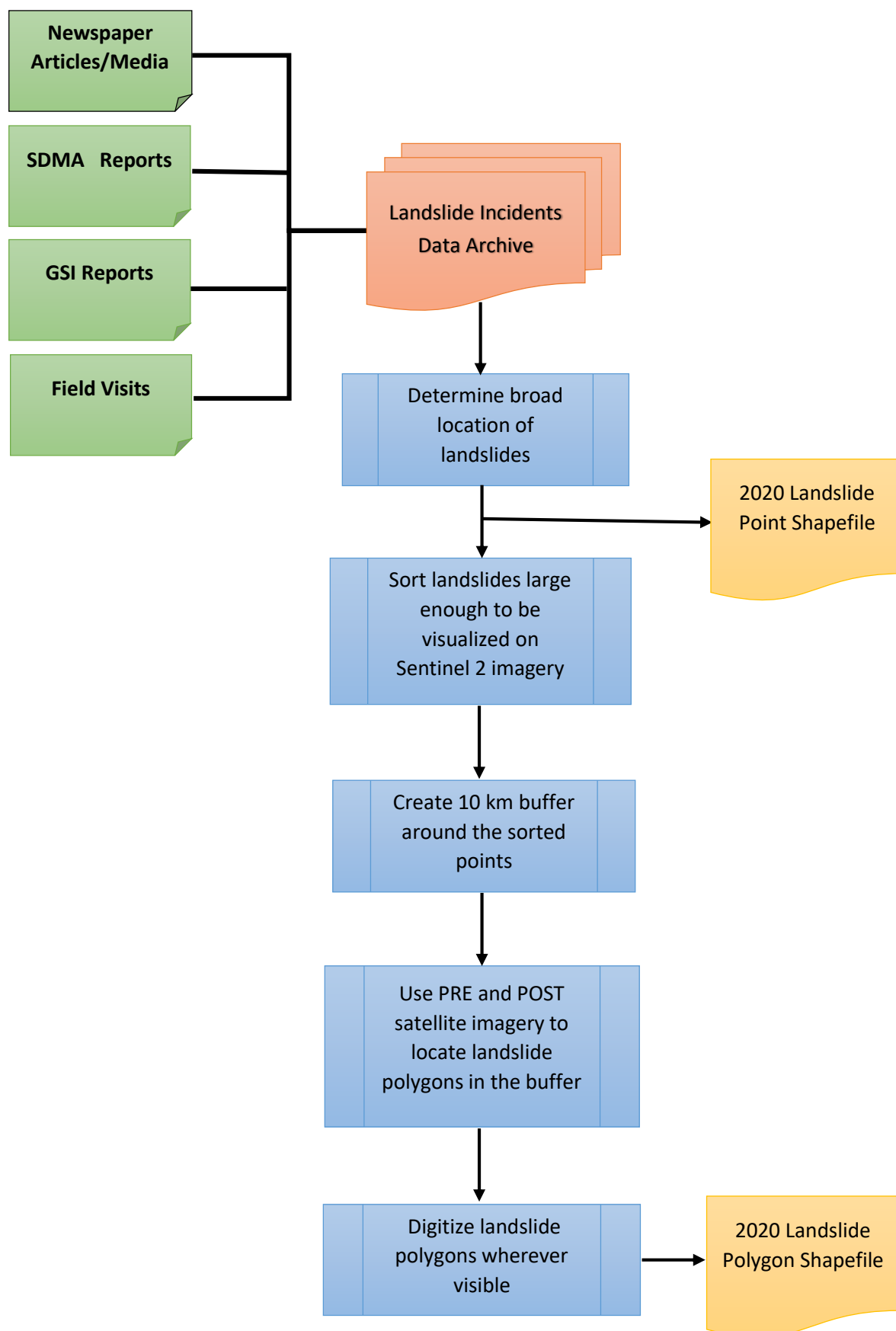


Figure 6: Flowchart for landslide polygons extraction

4.2 AUTOMATIC RAINFALL DATA EXTRACTION

From the list of landslides events, latitude, longitude and date of event are used to extract the corresponding rainfall data using our created automation pipeline. This automation pipeline will first download the required rainfall data from specific sources (Currently GPM data supported) and pre-process them to get hourly precipitation and cumulative precipitation of each event and finally generate the rainfall intensity graph for each event.

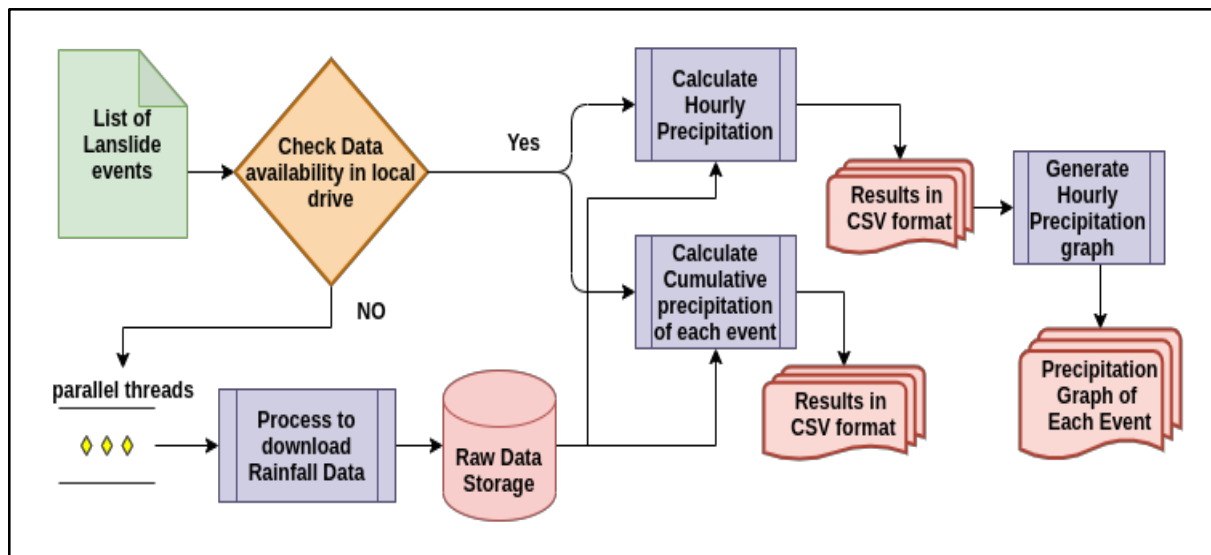


Figure 7: Flowchart for automatic extraction of rainfall

5. LANDSLIDE INVENTORY IN DIFFERENT PARTS OF THE NER

In this section of the report, a very brief summary of landslide incidents in the different states of the NER is provided. Detailed information regarding each landslide incident is provided in the annexure.

5.1 ARUNACHAL PRADESH

- There were a total of 5 landslide incidents recorded in Arunachal Pradesh.
- These landslide incidents lead to a total of 12 casualties in the state.

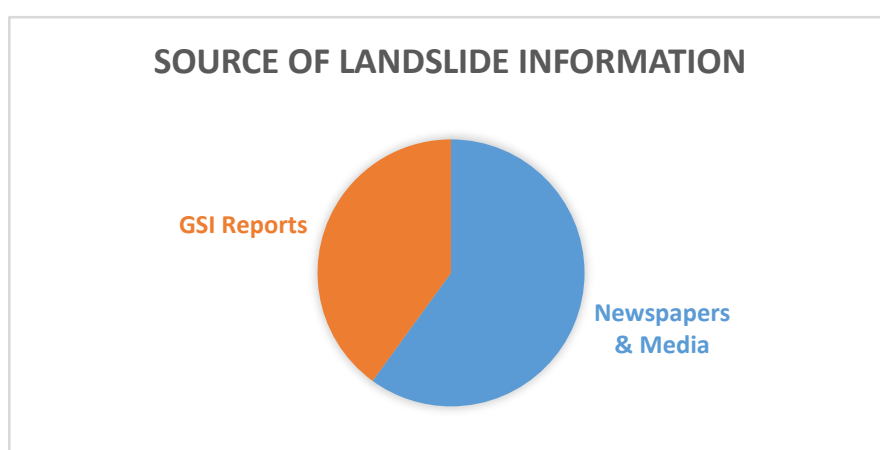
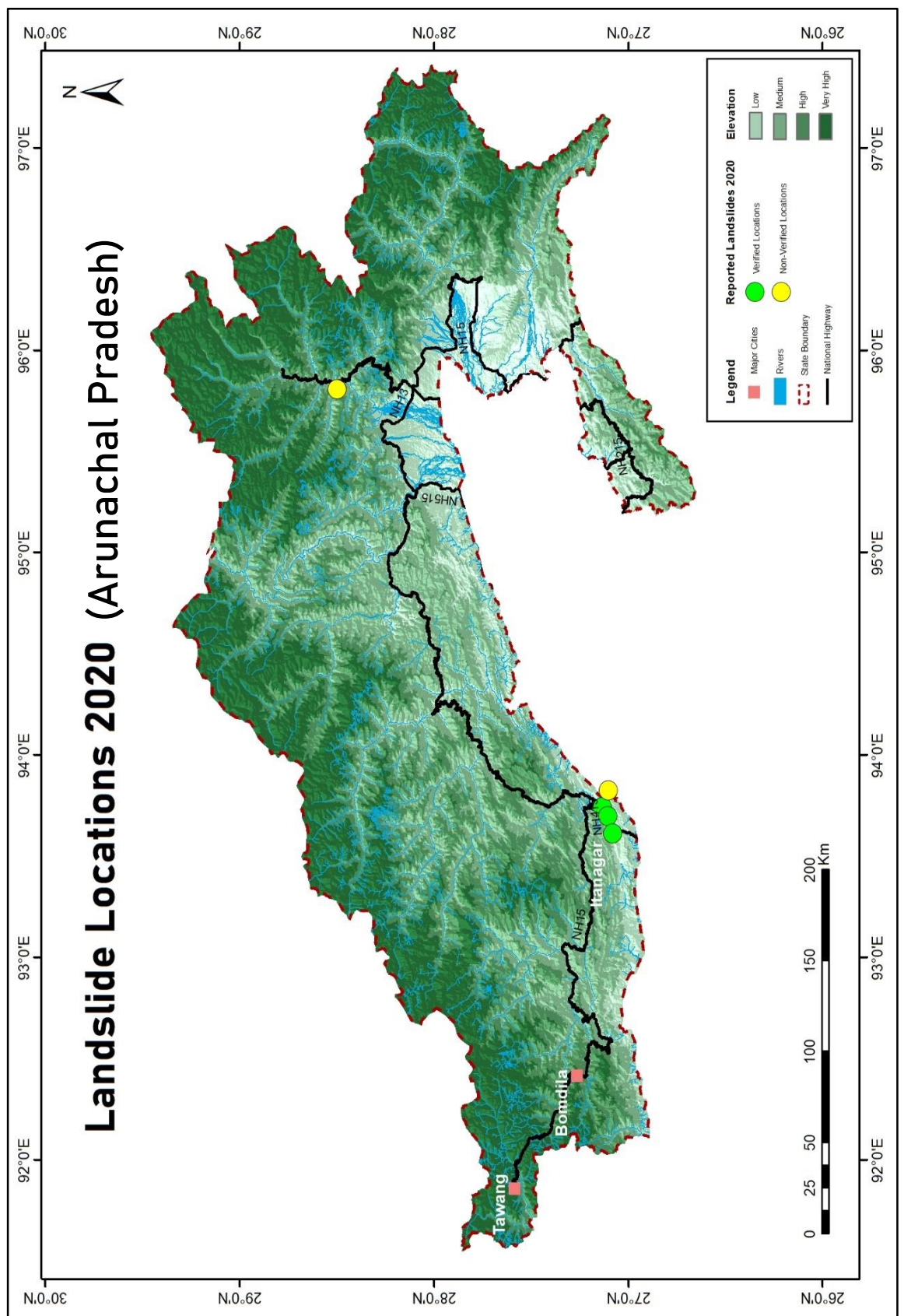


Figure 8a: Landslide Incidents reported in Arunachal Pradesh

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	GSI_REPORTS/2020/Arunachal Pradesh/21	GSI_REPORTS	Arunachal Pradesh	Papum Pare	10-Jul-20	1	4 dead
2	Newspaper&Media/2020/Arunachal Pradesh/22	Newspaper&Media	Arunachal Pradesh	Papum Pare	10-Jul-20	1	4 dead
3	GSI_REPORTS/2020/Arunachal Pradesh/23	GSI_REPORTS	Arunachal Pradesh	Papum Pare	25-Jun-20	1	1 dead
4	Newspaper&Media/2020/Arunachal Pradesh/24	Newspaper&Media	Arunachal Pradesh	Dibang Valley	25-May-20	2	3 dead
5	Newspaper&Media/2020/Arunachal Pradesh/25	Newspaper&Media	Arunachal Pradesh	Papum Pare	17-Sep-20	2	-

Figure 8b: Summary of landslide incidents in Arunachal Pradesh



Map 2: Landslide Incidents in Arunachal Pradesh 2020

5.2 ASSAM

- There were a total of 20 landslide incidences documented in Assam.
- There were a total of 22 deaths due to landslides in Assam of which 21 happened on the same day i.e. – 2nd June, 2020 in the Barak Valley region of the state.

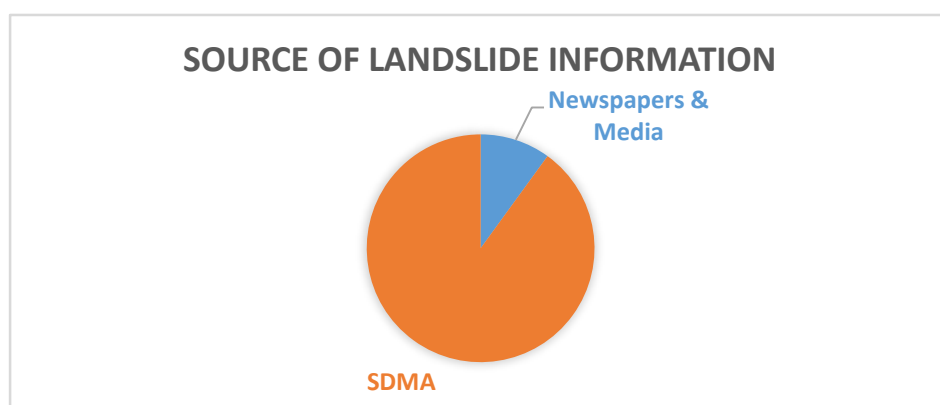
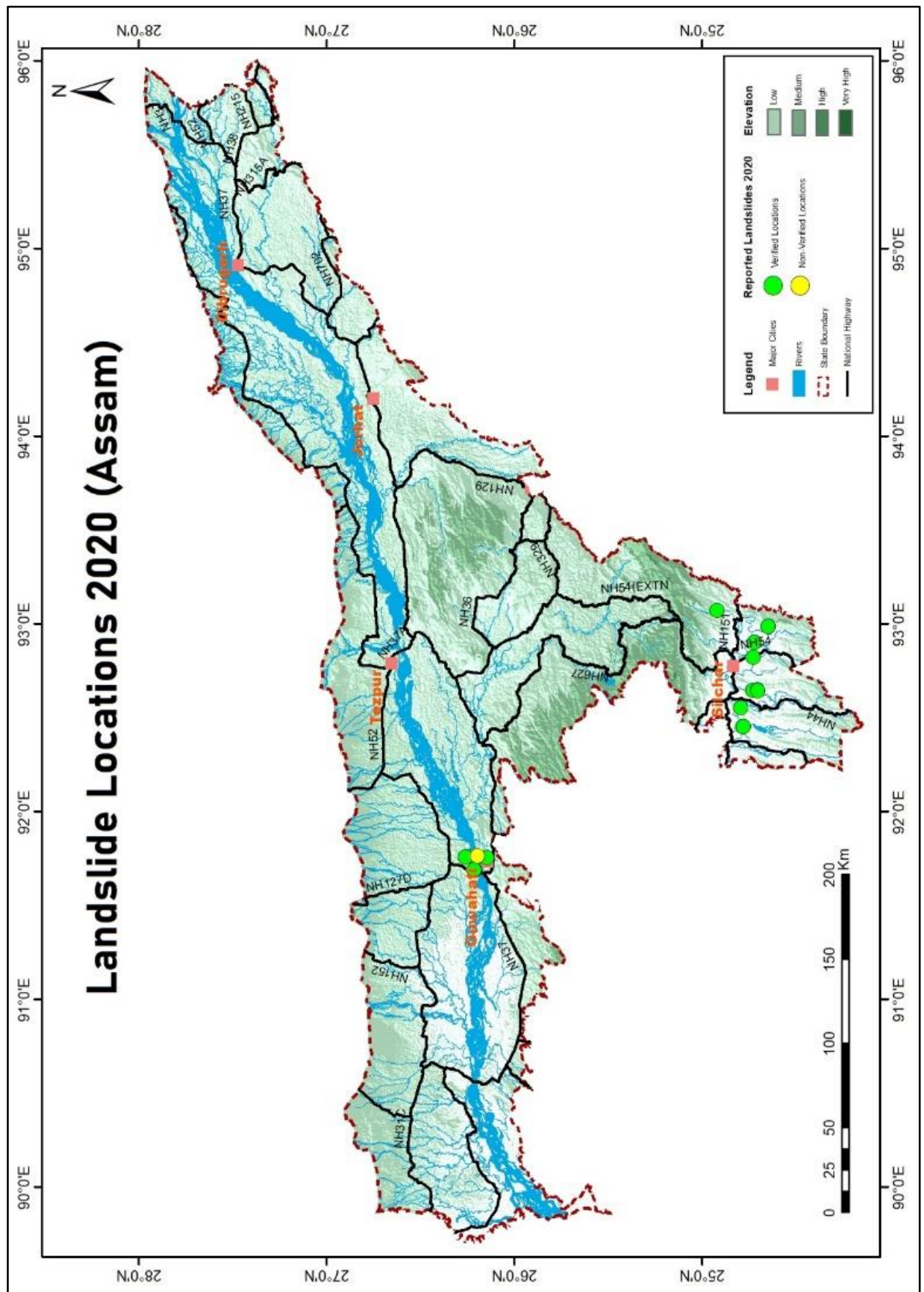


Figure 8a: Sources of landslide incidents reported from Assam

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	SDMA/2020/Assam/1	SDMA	Assam	Hailakandi	02.06.2020	1	1 dead
2	SDMA/2020/Assam/2	SDMA	Assam	Hailakandi	02.06.2020	1	1 dead
3	SDMA/2020/Assam/3	SDMA	Assam	Hailakandi	02.06.2020	1	1 dead
4	SDMA/2020/Assam/4	SDMA	Assam	Cachar	02.06.2020	1	7 dead
5	SDMA/2020/Assam/5	SDMA	Assam	Cachar	05.06.2020	1	-
6	SDMA/2020/Assam/6	SDMA	Assam	Hailakandi	02.06.2020	1	5 dead
7	SDMA/2020/Assam/7	SDMA	Assam	Karimganj	02.06.2020	1	1 dead
8	SDMA/2020/Assam/8	SDMA	Assam	Karimganj	02.06.2020	1	5 dead
9	SDMA/2020/Assam/9	SDMA	Assam	Kamrup metro	26.06.2020	1	1 dead
10	SDMA/2020/Assam/10	SDMA	Assam	Kamrup Rural	25.06.2020	1	-
11	SDMA/2020/Assam/11	SDMA	Assam	Cachar	08.06.2020	1	-
12	SDMA/2020/Assam/12	SDMA	Assam	Cachar	08.06.2020	1	-
13	SDMA/2020/Assam/13	SDMA	Assam	Cachar	08.06.2020	1	-
14	SDMA/2020/Assam/14	SDMA	Assam	Cachar	08.06.2020	1	-
15	SDMA/2020/Assam/15	SDMA	Assam	Kamrup Rural	26.06.2020	1	-
16	SDMA/2020/Assam/16	SDMA	Assam	Kamrup Rural	27.06.2020	1	-
17	SDMA/2020/Assam/17	SDMA	Assam	Kamrup Rural	27.06.2020	1	-
18	SDMA/2020/Assam/18	SDMA	Assam	Cachar	04.06.2020	1	-
19	Newspaper&Media/2020/Assam/19	Newspaper&Media	Assam	Kamrup metro	12.07.2020	1	2 dead
20	Newspaper&Media/2020/Assam/20	Newspaper&Media	Assam	Kamrup metro	28.06.2020	2	1 dead

Figure 9b: Summary of landslide incidents in Assam



Map 3: Landslide Incidents in Assam 2020

5.3 MANIPUR

- There were a total of 15 landslide incidences documented in Manipur in 2020.
- Several sections of the NH 37 and NH 2 were blocked due to landslides on 2nd June, 20th August and 23rd October.

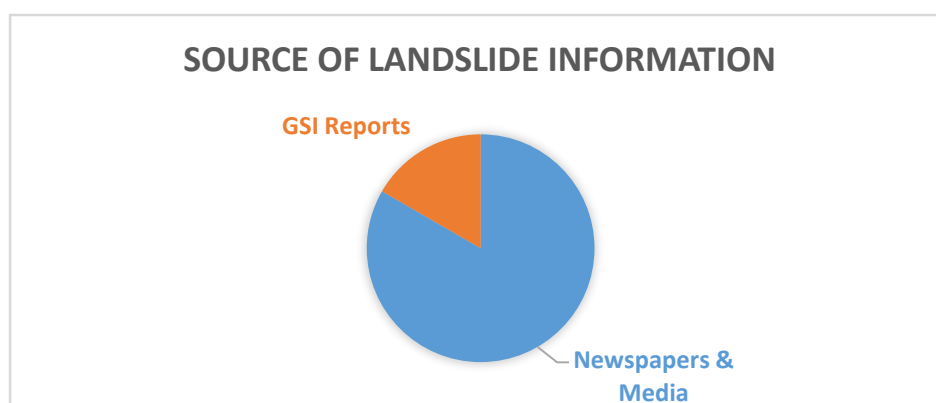
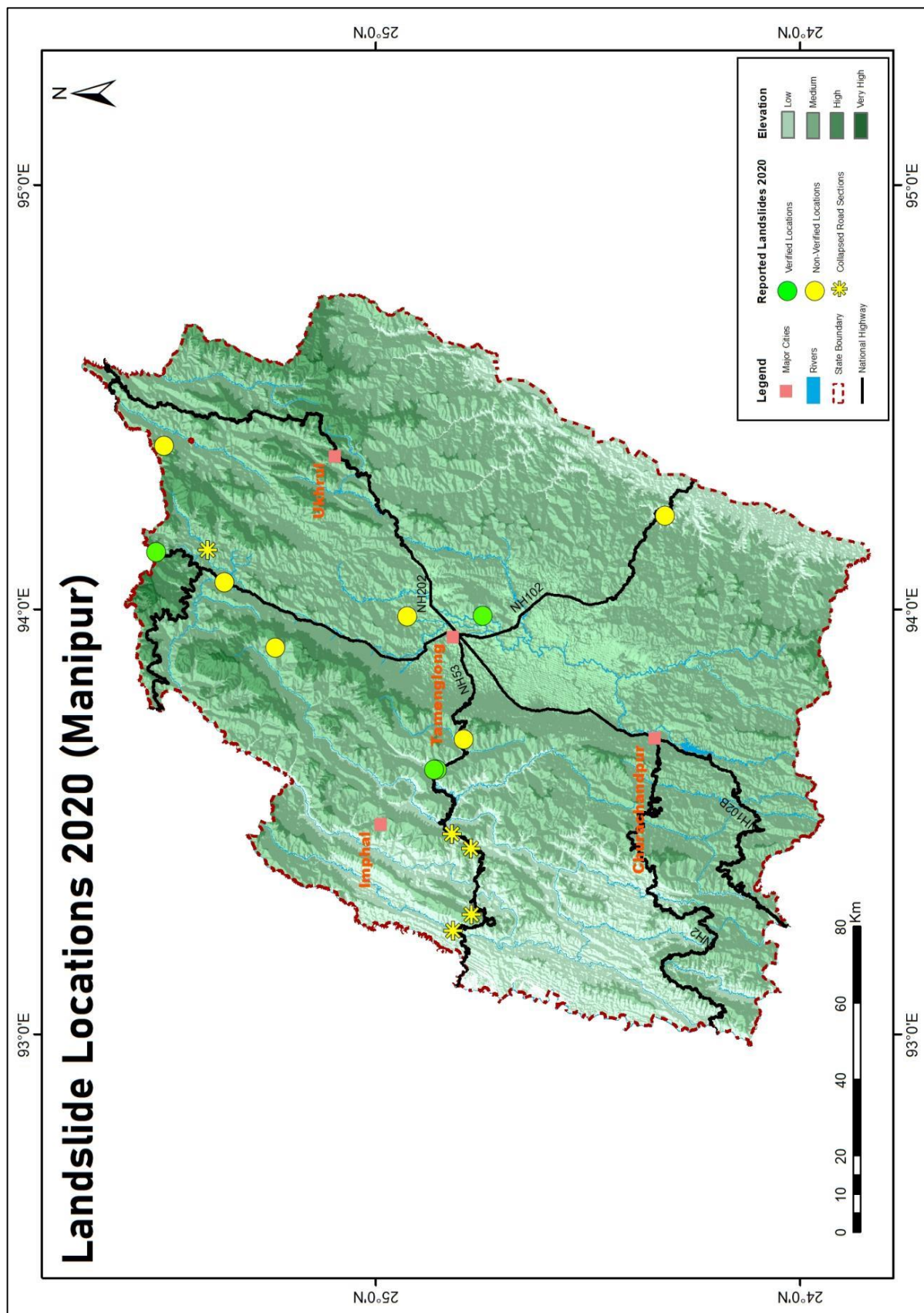


Figure 10a: Sources of landslide incidents reported from Manipur

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspaper&Media/2020/Manipur/26	Newspaper&Media	Manipur	Tamenglong	2-Jun-20	3	-
2	Newspaper&Media/2020/Manipur/27	Newspaper&Media	Manipur	Tamenglong	2-Jun-20	3	-
3	Newspaper&Media/2020/Manipur/28	Newspaper&Media	Manipur	Tamenglong	2-Jun-20	3	-
4	Newspaper&Media/2020/Manipur/29	Newspaper&Media	Manipur	Senapati	2-Jun-20	3	-
5	Newspaper&Media/2020/Manipur/30	Newspaper&Media	Manipur	Senapati	6-Jul-20	2	-
6	Newspaper&Media/2020/Manipur/31	Newspaper&Media	Manipur	Imphal east	11-Aug-20	1	-
7	Newspaper&Media/2020/Manipur/32	Newspaper&Media	Manipur	Senapati	21-Jun-20	2	-
8	Newspaper&Media/2020/Manipur/33	Newspaper&Media	Manipur	Senapati	20-Aug-20	2	-
9	Newspaper&Media/2020/Manipur/34	Newspaper&Media	Manipur	Senapati	28-Aug-20	2	-
10	Newspaper&Media/2020/Manipur/35	Newspaper&Media	Manipur	Chandel	20-Aug-20	2	-
11	Newspaper&Media/2020/Manipur/36	Newspaper&Media	Manipur	Tamenglong	27-May-20	3	-
12	Newspaper&Media/2020/Manipur/37	Newspaper&Media	Manipur	Senapati	23-Oct-20	2	-
13	GSI_REPORTS/2020/Manipur/38	GSI_REPORTS	Manipur	Tamenglong	Last week of June	1	-
14	GSI_REPORTS/2020/Manipur/39	GSI_REPORTS	Manipur	Tamenglong	Last week of June	1	-
15	GSI_REPORTS/2020/Manipur/40	GSI_REPORTS	Manipur	Senapati	Sometime before 21 July	1	-

Figure 10b: Summary of landslide incidents in Manipur



Map 4: Landslide Incidents in Manipur 2020

5.4 MEGHALAYA

- There were 23 landslide incidences that were documented in Meghalaya.
- A Field Visit was carried out to the site of the Thangshalai Landslide that occurred on 27 September 2020.

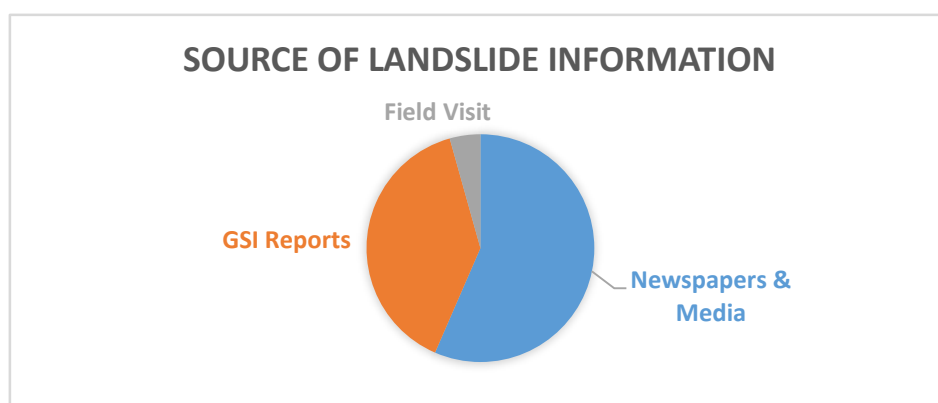
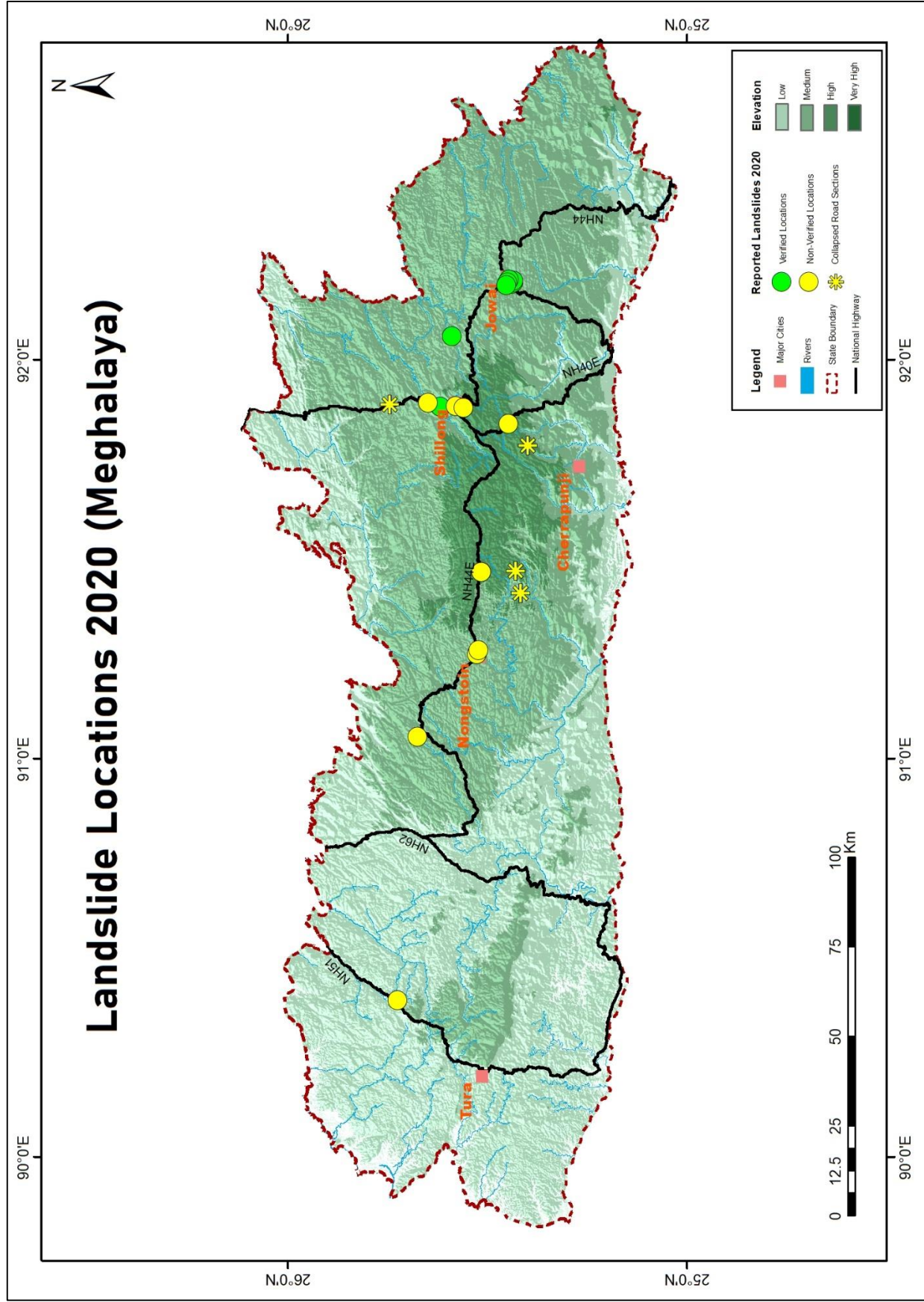


Figure 11a: Landslide Incidents reported from Meghalaya

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	GSI_REPORTS/2020/Meghalaya/41	GSI_REPORTS	Meghalaya	East Khasi Hills	21-May-20	1	-
2	GSI_REPORTS/2020/Meghalaya/42	GSI_REPORTS	Meghalaya	Jaintia Hills	26-May-20	1	-
3	GSI_REPORTS/2020/Meghalaya/43	GSI_REPORTS	Meghalaya	Jaintia Hills	26-May-20	1	-
4	GSI_REPORTS/2020/Meghalaya/44	GSI_REPORTS	Meghalaya	Jaintia Hills	26-May-20	1	-
5	GSI_REPORTS/2020/Meghalaya/45	GSI_REPORTS	Meghalaya	Jaintia Hills	26-May-20	1	-
6	GSI_REPORTS/2020/Meghalaya/46	GSI_REPORTS	Meghalaya	Jaintia Hills	26-May-20	1	-
7	GSI_REPORTS/2020/Meghalaya/47	GSI_REPORTS	Meghalaya	Jaintia Hills	26-May-20	1	-
8	GSI_REPORTS/2020/Meghalaya/48	GSI_REPORTS	Meghalaya	Jaintia Hills	26-May-20	1	-
9	GSI_REPORTS/2020/Meghalaya/49	GSI_REPORTS	Meghalaya	Jaintia Hills	26-May-20	1	-
10	Newspaper&Media/2020/Meghalaya/50	Newspaper&Media	Meghalaya	East Khasi Hills	11-Jul-20	2	-
11	FIELD_VISIT/2020/Meghalaya/51	FIELD_VISIT	Meghalaya	East Khasi Hills	27-Sep-20	1	-
12	Newspaper&Media/2020/Meghalaya/52	Newspaper&Media	Meghalaya	Ri-bhoi	24-Oct-20	3	-
13	Newspaper&Media/2020/Meghalaya/53	Newspaper&Media	Meghalaya	West Khasi Hills	10-Jul-20	3	-
14	Newspaper&Media/2020/Meghalaya/54	Newspaper&Media	Meghalaya	West Khasi Hills	9-Jul-20	3	-
15	Newspaper&Media/2020/Meghalaya/55	Newspaper&Media	Meghalaya	West Khasi Hills	11-Jul-20	2	-
16	Newspaper&Media/2020/Meghalaya/56	Newspaper&Media	Meghalaya	East Khasi Hills	12-Jul-20	2	2 dead
17	Newspaper&Media/2020/Meghalaya/57	Newspaper&Media	Meghalaya	East Khasi Hills	25-Sep-20	2	5 dead
18	Newspaper&Media/2020/Meghalaya/58	Newspaper&Media	Meghalaya	West Garo Hills	24-Jul-20	2	-
19	Newspaper&Media/2020/Meghalaya/59	Newspaper&Media	Meghalaya	West Khasi Hills	14-Jul-20	2	-
20	Newspaper&Media/2020/Meghalaya/60	Newspaper&Media	Meghalaya	West Khasi Hills	14-Jul-20	2	-
21	Newspaper&Media/2020/Meghalaya/61	Newspaper&Media	Meghalaya	West Khasi Hills	14-Jul-20	2	-
22	Newspaper&Media/2020/Meghalaya/62	Newspaper&Media	Meghalaya	East Khasi Hills	24-Sep-20	2	-
23	Newspaper&Media/2020/Meghalaya/63	Newspaper&Media	Meghalaya	East Khasi Hills	29-May-20	3	-

Figure 11b: Summary of landslide incidents from Meghalaya



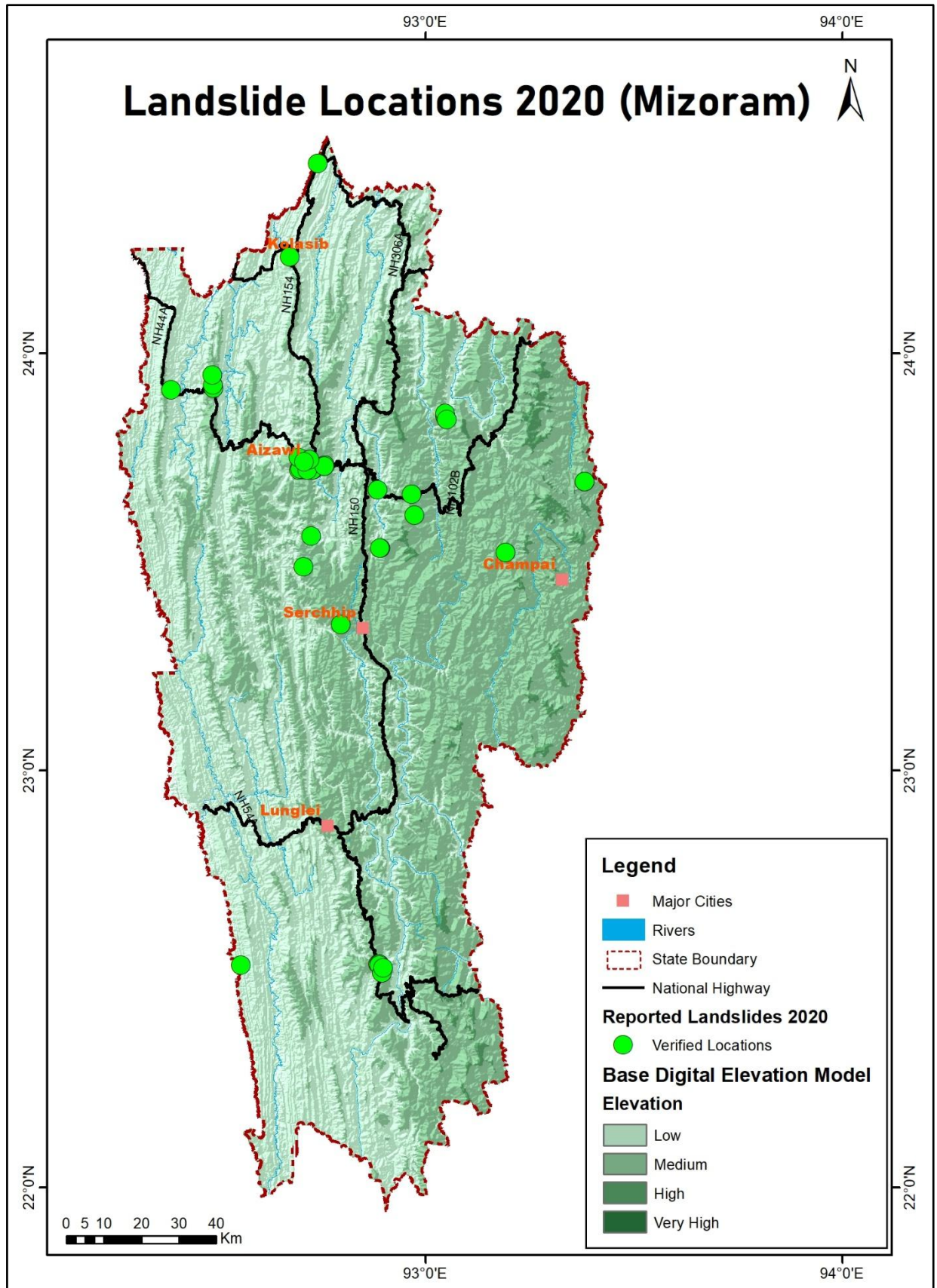
Map 5: Landslide Incidents in Meghalaya 2020

5.5 MIZORAM

- Total of 41 landslide incidents occurred in Mizoram
- All of these landslide incidents could be verified from SDMA Reports.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	SDMA/2020/Mizoram/64	SDMA	Mizoram	Aizawl	06.10.2020	1	-
2	SDMA/2020/Mizoram/65	SDMA	Mizoram	Aizawl	22.09.2020	1	-
3	SDMA/2020/Mizoram/66	SDMA	Mizoram	Aizawl	01.10.2020	1	-
4	SDMA/2020/Mizoram/67	SDMA	Mizoram	Aizawl	05.06.2020	1	-
5	SDMA/2020/Mizoram/68	SDMA	Mizoram	Aizawl	27.09.2020	1	-
6	SDMA/2020/Mizoram/69	SDMA	Mizoram	Aizawl	13.08.2020	1	-
8	SDMA/2020/Mizoram/71	SDMA	Mizoram	Aizawl	02.10.2020	1	-
9	SDMA/2020/Mizoram/72	SDMA	Mizoram	Aizawl	04.07.2020	1	-
10	SDMA/2020/Mizoram/73	SDMA	Mizoram	Aizawl	31.08.2020	1	-
11	SDMA/2020/Mizoram/74	SDMA	Mizoram	Aizawl	10.06.2020	1	-
12	SDMA/2020/Mizoram/75	SDMA	Mizoram	Aizawl	11.07.2020	1	-
13	SDMA/2020/Mizoram/76	SDMA	Mizoram	Aizawl	02.02.2020	1	-
14	SDMA/2020/Mizoram/77	SDMA	Mizoram	Aizawl	15.06.2020	1	-
15	SDMA/2020/Mizoram/78	SDMA	Mizoram	Aizawl	17.07.2020	1	-
16	SDMA/2020/Mizoram/79	SDMA	Mizoram	Aizawl	28.05.2020	1	-
17	SDMA/2020/Mizoram/80	SDMA	Mizoram	Aizawl	10.07.2020	1	-
18	SDMA/2020/Mizoram/81	SDMA	Mizoram	Champhai	08.08.2020	1	-
19	SDMA/2020/Mizoram/82	SDMA	Mizoram	Champhai	01.08.2020	1	-
20	SDMA/2020/Mizoram/83	SDMA	Mizoram	Lawngtlai	07.07.2020	1	-
21	SDMA/2020/Mizoram/84	SDMA	Mizoram	Lawngtlai	03.08.2020	1	-
22	SDMA/2020/Mizoram/85	SDMA	Mizoram	Serchhip	04.08.2020	1	-
23	SDMA/2020/Mizoram/86	SDMA	Mizoram	Lawngtlai	03.08.2020	1	-
24	SDMA/2020/Mizoram/87	SDMA	Mizoram	Lawngtlai	20.08.2020	1	-
25	SDMA/2020/Mizoram/88	SDMA	Mizoram	Serchhip	24.08.2020	1	-
26	SDMA/2020/Mizoram/89	SDMA	Mizoram	Lawngtlai	02.08.2020	1	-
27	SDMA/2020/Mizoram/90	SDMA	Mizoram	Lawngtlai	03.08.2020	1	-
28	SDMA/2020/Mizoram/91	SDMA	Mizoram	Aizawl	02.10.2020	1	-
29	SDMA/2020/Mizoram/92	SDMA	Mizoram	Aizawl	02.10.2020 – 03.10.2020	1	-
30	SDMA/2020/Mizoram/93	SDMA	Mizoram	Aizawl	17.08.2020	1	-
31	SDMA/2020/Mizoram/94	SDMA	Mizoram	Aizawl	13.10.2020	1	-
32	SDMA/2020/Mizoram/95	SDMA	Mizoram	Dhalai*	08.10.2020	1	-
33	SDMA/2020/Mizoram/96	SDMA	Mizoram	Aizawl	10.07.2020	1	-
34	SDMA/2020/Mizoram/97	SDMA	Mizoram	Mamit	10.06.2020	1	-
35	SDMA/2020/Mizoram/98	SDMA	Mizoram	Mamit	11.07.2020	1	-
36	SDMA/2020/Mizoram/99	SDMA	Mizoram	Mamit	11.06.2020	1	-
37	SDMA/2020/Mizoram/100	SDMA	Mizoram	Mamit	11.06.2020	1	-
38	SDMA/2020/Mizoram/101	SDMA	Mizoram	Mamit	29.05.2020	1	-
39	SDMA/2020/Mizoram/102	SDMA	Mizoram	Champhai	03.10.2020	1	-
40	SDMA/2020/Mizoram/103	SDMA	Mizoram	Kolasib	21.10.2020	1	-
41	SDMA/2020/Mizoram/104	SDMA	Mizoram	Kolasib	25.05.2020	1	-

Figure 12: Summary of landslide incidents from Mizoram



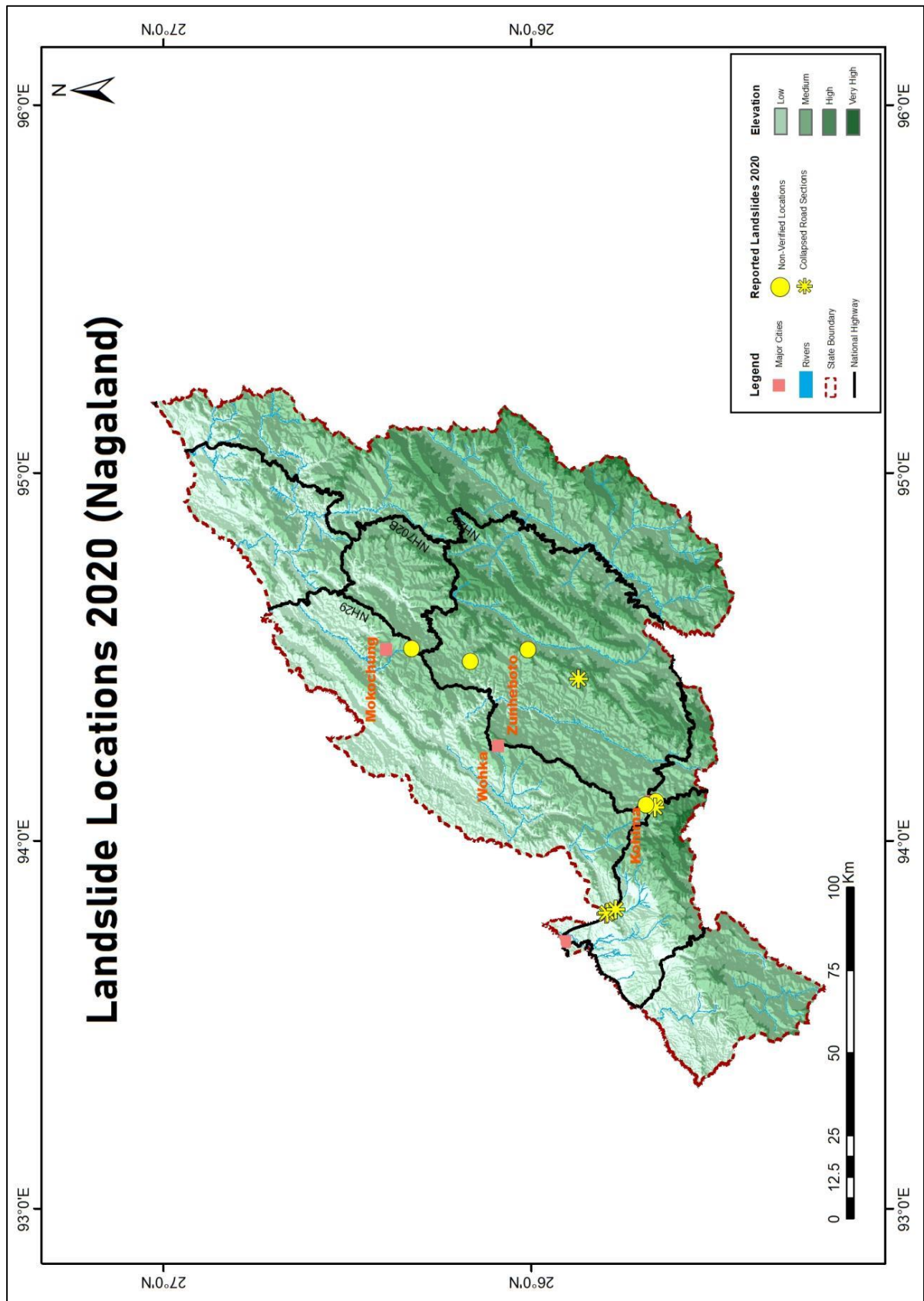
Map 6: Landslide Incidents in Mizoram 2020

5.6 NAGALAND

- All of the 10 landslide incidents in Nagaland were sourced from Newspapers and Media.
- The landslide that occurred on 8th July 2020 in the Zunheboto district claimed the life of 1 person.
- The construction of a 4 lane highway along NH 29, led to blocking of several sections of the NH 29 near Kukidalong and Chumoukedima Check Gate due to landslides on 7th and 19th September 2020.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspaper&Media/2020/Nagaland/105	Newspaper&Media	Nagaland	Kohima	4-Oct-20	2	-
2	Newspaper&Media/2020/Nagaland/106	Newspaper&Media	Nagaland	Mokokchung	15-Aug-20	2	-
3	Newspaper&Media/2020/Nagaland/107	Newspaper&Media	Nagaland	Zunheboto	17-Aug-20	2	-
4	Newspaper&Media/2020/Nagaland/108	Newspaper&Media	Nagaland	Zunheboto	14-Aug-20	2	-
5	Newspaper&Media/2020/Nagaland/109	Newspaper&Media	Nagaland	Dimapur	7-Sep-20	3	-
6	Newspaper&Media/2020/Nagaland/110	Newspaper&Media	Nagaland	Dimapur	12-Sep-20	3	-
7	Newspaper&Media/2020/Nagaland/111	Newspaper&Media	Nagaland	Dimapur	19-Sep-20	3	-
8	Newspaper&Media/2020/Nagaland/112	Newspaper&Media	Nagaland	Kohima	19-Aug-20	3	-
9	Newspaper&Media/2020/Nagaland/113	Newspaper&Media	Nagaland	Zunheboto	8-Jul-20	3	1 dead
10	Newspaper&Media/2020/Nagaland/114	Newspaper&Media	Nagaland	Kohima	11-Jul-20	2	-

Figure 13: Summary of Landslide Incidents from Nagaland



Map 7: Landslide Incidents in Nagaland 2020

5.7 SIKKIM

- A total of 27 landslide incidents were documented in the year 2020 for Sikkim.
- There were a total of 7 casualties reported from Sikkim.

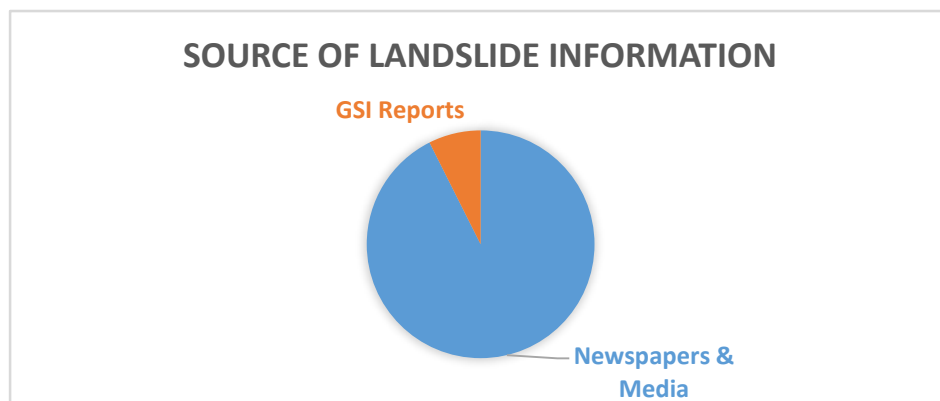
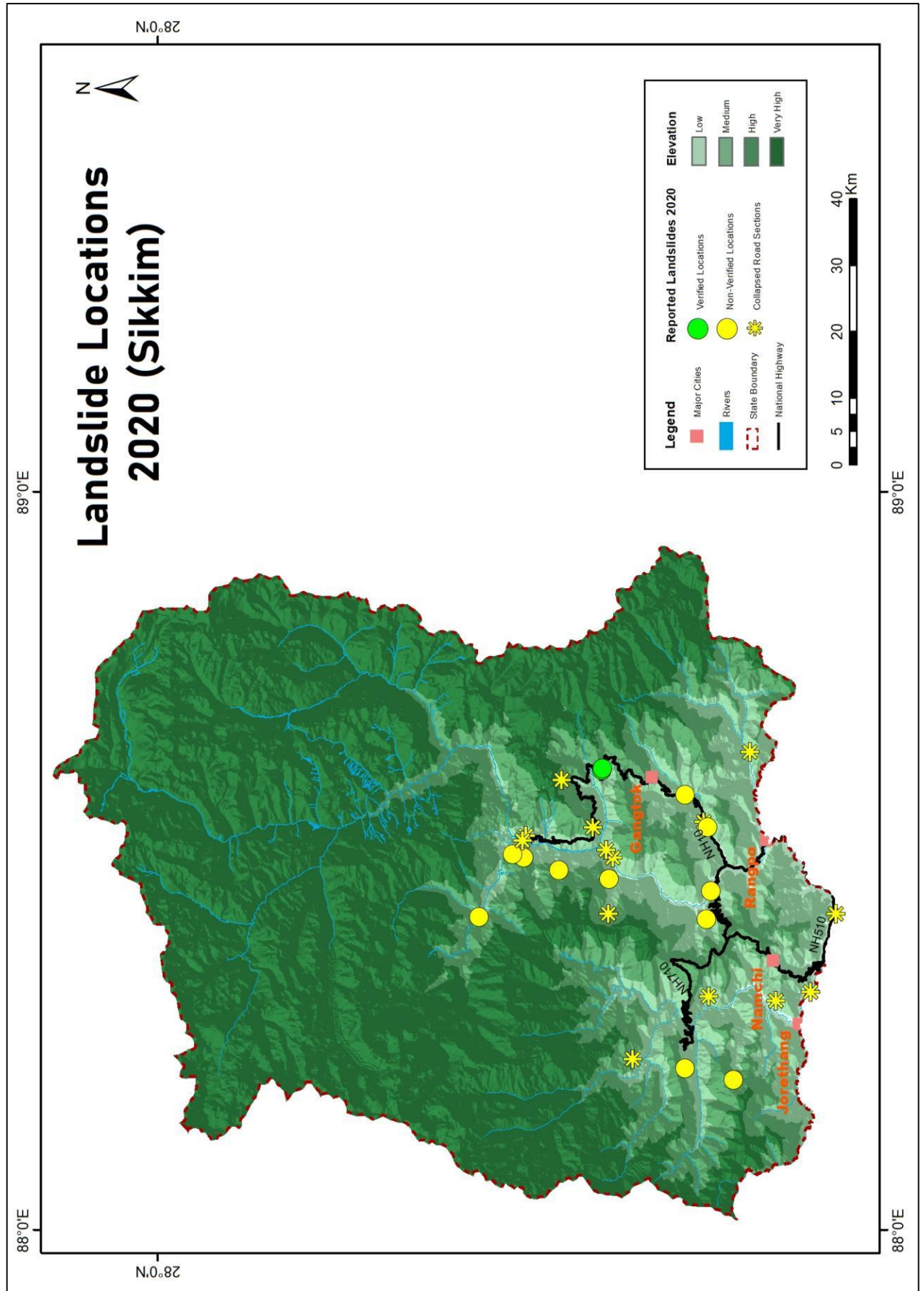


Figure 14a: Landslide Incidents reported from Sikkim

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspaper&Media/2020/Sikkim/115	Newspaper&Media	Sikkim	West Sikkim	24-May-20	2	1 dead
2	Newspaper&Media/2020/Sikkim/116	Newspaper&Media	Sikkim	East Sikkim	8-Sep-20	3	-
3	Newspaper&Media/2020/Sikkim/117	Newspaper&Media	Sikkim	East Sikkim	8-Sep-20	2	-
4	Newspaper&Media/2020/Sikkim/118	Newspaper&Media	Sikkim	East Sikkim	8-Sep-20	2	-
5	Newspaper&Media/2020/Sikkim/119	Newspaper&Media	Sikkim	South Sikkim	8-Jun-20	2	2 dead
6	Newspaper&Media/2020/Sikkim/120	Newspaper&Media	Sikkim	West Sikkim	2-Oct-20	2	2 dead
7	Newspaper&Media/2020/Sikkim/121	Newspaper&Media	Sikkim	North Sikkim	29-Jun-20	3	-
8	Newspaper&Media/2020/Sikkim/122	Newspaper&Media	Sikkim	North Sikkim	27-Jun-20	2	-
9	Newspaper&Media/2020/Sikkim/123	Newspaper&Media	Sikkim	North Sikkim	27-Jun-20	2	-
10	Newspaper&Media/2020/Sikkim/124	Newspaper&Media	Sikkim	North Sikkim	27-Jun-20	2	-
11	Newspaper&Media/2020/Sikkim/125	Newspaper&Media	Sikkim	North Sikkim	27-Jun-20	2	-
12	Newspaper&Media/2020/Sikkim/126	Newspaper&Media	Sikkim	North Sikkim	27-Jun-20	3	-
13	Newspaper&Media/2020/Sikkim/127	Newspaper&Media	Sikkim	West Sikkim	28-Jun-20	3	-
14	Newspaper&Media/2020/Sikkim/128	Newspaper&Media	Sikkim	East Sikkim	27-Jun-20	2	-
15	Newspaper&Media/2020/Sikkim/129	Newspaper&Media	Sikkim	North Sikkim	26-May-20	2	2 dead
16	Newspaper&Media/2020/Sikkim/130	Newspaper&Media	Sikkim	North Sikkim	25-Jun-20	3	-
17	Newspaper&Media/2020/Sikkim/131	Newspaper&Media	Sikkim	North Sikkim	25-Jun-20	3	-
18	Newspaper&Media/2020/Sikkim/132	Newspaper&Media	Sikkim	East Sikkim	25-Jun-20	3	-
19	GSI_REPORTS/2020/Sikkim/133	GSI_REPORTS	Sikkim	North Sikkim	24-May-20	1	-
20	GSI_REPORTS/2020/Sikkim/134	GSI_REPORTS	Sikkim	North Sikkim	24-May-20	1	-
21	Newspaper&Media/2020/Sikkim/135	Newspaper&Media	Sikkim	South Sikkim	1-Jun-20	3	-
22	Newspaper&Media/2020/Sikkim/136	Newspaper&Media	Sikkim	South Sikkim	6-Aug-20	3	-
23	Newspaper&Media/2020/Sikkim/137	Newspaper&Media	Sikkim	East Sikkim	8 to 11 September 2020	3	-
24	Newspaper&Media/2020/Sikkim/138	Newspaper&Media	Sikkim	East Sikkim	8 to 11 September 2020	3	-
25	Newspaper&Media/2020/Sikkim/139	Newspaper&Media	Sikkim	South Sikkim	8 to 11 September 2020	3	-
26	Newspaper&Media/2020/Sikkim/140	Newspaper&Media	Sikkim	South Sikkim	8 to 11 September 2020	3	-
27	Newspaper&Media/2020/Sikkim/141	Newspaper&Media	Sikkim	South Sikkim	8 to 11 September 2020	3	-

Figure 14b: Summary of Landslide Incidents from Sikkim



Map 8: Landslide Incidents in Sikkim 2020

6. ANNEXURES

ANNEXURE I: STATEWISE LIST OF LANDSLIDE INCIDENTS IN NER

ARUNACHAL PRADESH

Slide ID	GSI_REPORTS/2020/Arunachal Pradesh/21
Source	GSI_REPORTS
Slide No	21
State	Arunachal Pradesh
District	Papum Pare
Name	Tigdo Landslide
Location	Yupia ? Potin Road, about 4 km from Doimukh town
NH-SH Cutoff/Affected	
Lat (dd)	27.134694
Long (dd)	93.743472
Date of Event	10-Jul-20
Time of Event	2:30 A.M
Type of Event	1
Casualty	4 dead
Rainfall (mm)	46.26270328

Slide ID	Newspaper&Media/2020/Arunachal Pradesh/22
Source	Newspaper&Media
Slide No	22
State	Arunachal Pradesh
District	Papum Pare
Name	Modrijo
Location	Itanagar and Naharlagun
NH-SH Cutoff/Affected	
Lat (dd)	27.107
Long (dd)	93.698
Date of Event	10-Jul-20
Time of Event	
Type of Event	1
Casualty	4 dead
Rainfall (mm)	88.02032345

Slide ID	GSI_REPORTS/2020/Arunachal Pradesh/23
Source	GSI_REPORTS
Slide No	23
State	Arunachal Pradesh
District	Papum Pare
Name	Donyi Colony
Location	Donyi Colony Near Jollang Road, 2.5 km east of Itanagar City
NH-SH Cutoff/Affected	
Lat (dd)	27.081306
Long (dd)	93.612833
Date of Event	25-Jun-20
Time of Event	8:12 A.M
Type of Event	1
Casualty	1 dead
Rainfall (mm)	27.01856839

Slide ID	Newspaper&Media/2020/Arunachal Pradesh/24
Source	Newspaper&Media
Slide No	24
State	Arunachal Pradesh
District	Dibang Valley
Name	Aarzo Village
Location	Aarzo Village Dibang Valley district
NH-SH Cutoff/Affected	
Lat (dd)	28.5
Long (dd)	95.807
Date of Event	25-May-20
Time of Event	
Type of Event	2
Casualty	3 dead
Rainfall (mm)	45.72257885

Slide ID	Newspaper&Media/2020/Arunachal Pradesh/25
Source	Newspaper&Media
Slide No	25
State	Arunachal Pradesh
District	Papum Pare
Name	Bandardewa Landslide
Location	Bandardewa area at NH 15
NH-SH Cutoff/Affected	NH 15
Lat (dd)	27.104
Long (dd)	93.826
Date of Event	17-Sep-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	10.09935397

ASSAM

Slide ID	SDMA/2020/Assam/1
Source	SDMA
Slide No	1
State	Assam
District	Hailakandi
Name	SDMA/AS/2020/Chandipur Grant Slide
Location	Chandipur, Bowarhar, Algapur Circle, Hailakandi
NH-SH Cutoff/Affected	
Lat (dd)	24.79488889
Long (dd)	92.55505556
Date of Event	02.06.2020
Time of Event	
Type of Event	1
Casualty	1 dead
Rainfall (mm)	72.78251015

Slide ID	SDMA/2020/Assam/2
Source	SDMA
Slide No	2
State	Assam
District	Hailakandi
Name	SDMA/AS/2020/Mohanpur Grant Slide 3
Location	Mohanpur Grant, Hailakandi Circle, Hailakandi
NH-SH Cutoff/Affected	
Lat (dd)	24.73116667
Long (dd)	92.64738889
Date of Event	02.06.2020
Time of Event	
Type of Event	1
Casualty	1 dead
Rainfall (mm)	59.14748733

Slide ID	SDMA/2020/Assam/3
Source	SDMA
Slide No	3
State	Assam
District	Hailakandi
Name	SDMA/AS/2020/Mohanpur Grant Slide 2
Location	Mohanpur Grant, Hailakandi Circle, Hailakandi
NH-SH Cutoff/Affected	
Lat (dd)	24.70655556
Long (dd)	92.65008333
Date of Event	02.06.2020
Time of Event	
Type of Event	1
Casualty	1 dead
Rainfall (mm)	59.14748733

Slide ID	SDMA/2020/Assam/4
Source	SDMA
Slide No	4
State	Assam
District	Cachar
Name	SDMA/AS/2020/Kanakpur Slide
Location	Kanakpur Pt III, Lakhipur, Cachar
NH-SH Cutoff/Affected	
Lat (dd)	24.92097222
Long (dd)	93.07305556
Date of Event	02.06.2020
Time of Event	
Type of Event	1
Casualty	7 dead
Rainfall (mm)	29.16217088

Slide ID	SDMA/2020/Assam/5
Source	SDMA
Slide No	5
State	Assam
District	Cachar
Name	SDMA/AS/2020/Govindanagar Slide
Location	Govindanagar Pt. II, Sonai, Cachar
NH-SH Cutoff/Affected	
Lat (dd)	24.72191667
Long (dd)	92.90380556
Date of Event	05.06.2020
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	79.11487476

Slide ID	SDMA/2020/Assam/6
Source	SDMA
Slide No	6
State	Assam
District	Hailakandi
Name	SDMA/AS/2020/Mohanpur Grant Slide 1
Location	Mohanpur Grant, Hailakandi Circle, Hailakandi
NH-SH Cutoff/Affected	
Lat (dd)	24.70363889
Long (dd)	92.64680556
Date of Event	02.06.2020
Time of Event	
Type of Event	1
Casualty	5 dead
Rainfall (mm)	59.14748733

Slide ID	SDMA/2020/Assam/7
Source	SDMA
Slide No	7
State	Assam
District	Karimganj
Name	SDMA/AS/2020/Karimpur Slide 1
Location	Karimpur, Nilam Bazar, Karimganj
NH-SH Cutoff/Affected	
Lat (dd)	24.78380556
Long (dd)	92.45563889
Date of Event	02.06.2020
Time of Event	
Type of Event	1
Casualty	1 dead
Rainfall (mm)	62.41079903

Slide ID	SDMA/2020/Assam/8
Source	SDMA
Slide No	8
State	Assam
District	Karimganj
Name	SDMA/AS/2020/Karimpur Slide 2
Location	Karimpur, Nilam Bazar, Karimganj
NH-SH Cutoff/Affected	
Lat (dd)	24.78002778
Long (dd)	92.45372222
Date of Event	02.06.2020
Time of Event	
Type of Event	1
Casualty	5 dead
Rainfall (mm)	62.41079903

Slide ID	SDMA/2020/Assam/9
Source	SDMA
Slide No	9
State	Assam
District	Kamrup metro
Name	SDMA/AS/2020/Kahilipara Slide
Location	Power grid, Kahilipara, Guwahati
NH-SH Cutoff/Affected	
Lat (dd)	26.146
Long (dd)	91.756
Date of Event	26.06.2020
Time of Event	
Type of Event	1
Casualty	1 dead
Rainfall (mm)	196.6897646

Slide ID	SDMA/2020/Assam/10
Source	SDMA
Slide No	10
State	Assam
District	Kamrup Rural
Name	SDMA/AS/2020/Amingaon Slide 01
Location	Amingaon, Guwahati
NH-SH Cutoff/Affected	
Lat (dd)	26.21083333
Long (dd)	91.68944444
Date of Event	25.06.2020
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	58.38482125

Slide ID	SDMA/2020/Assam/11
Source	SDMA
Slide No	11
State	Assam
District	Cachar
Name	SDMA/AS/2020/SONAI1
Location	Panichowki & Saidpur 4
NH-SH Cutoff/Affected	
Lat (dd)	24.647875
Long (dd)	92.988353
Date of Event	08.06.2020
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	18.524112

Slide ID	SDMA/2020/Assam/12
Source	SDMA
Slide No	12
State	Assam
District	Cachar
Name	SDMA/AS/2020/SONAI2
Location	Panichowki & Saidpur 4
NH-SH Cutoff/Affected	
Lat (dd)	24.646983
Long (dd)	92.98905
Date of Event	08.06.2020
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	18.524112

Slide ID	SDMA/2020/Assam/13
Source	SDMA
Slide No	13
State	Assam
District	Cachar
Name	SDMA/AS/2020/SONAI3
Location	Panichowki & Saidpur 4
NH-SH Cutoff/Affected	
Lat (dd)	24.649889
Long (dd)	92.989075
Date of Event	08.06.2020
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	18.524112

Slide ID	SDMA/2020/Assam/14
Source	SDMA
Slide No	14
State	Assam
District	Cachar
Name	SDMA/AS/2020/SONAI4
Location	Panichowki & Saidpur 4
NH-SH Cutoff/Affected	
Lat (dd)	24.727878
Long (dd)	92.823175
Date of Event	08.06.2020
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	27.53328682

Slide ID	SDMA/2020/Assam/15
Source	SDMA
Slide No	15
State	Assam
District	Kamrup Rural
Name	SDMA/AS/2020/North Guwahati1
Location	Gauripur
NH-SH Cutoff/Affected	
Lat (dd)	26.214209
Long (dd)	91.693519
Date of Event	26.06.2020
Time of Event	4:00 P.M
Type of Event	1
Casualty	
Rainfall (mm)	92.36848196

Slide ID	SDMA/2020/Assam/16
Source	SDMA
Slide No	16
State	Assam
District	Kamrup Rural
Name	SDMA/AS/2020/North Guwahati2
Location	DIRGHESWARI
NH-SH Cutoff/Affected	
Lat (dd)	26.254368
Long (dd)	91.762678
Date of Event	27.06.2020
Time of Event	6:00 P.M
Type of Event	1
Casualty	
Rainfall (mm)	102.674153

Slide ID	SDMA/2020/Assam/17
Source	SDMA
Slide No	17
State	Assam
District	Kamrup Rural
Name	SDMA/AS/2020/North Guwahati3
Location	DIRGHESWARI
NH-SH Cutoff/Affected	
Lat (dd)	26.262201
Long (dd)	91.759778
Date of Event	27.06.2020
Time of Event	6:00 P.M
Type of Event	1
Casualty	
Rainfall (mm)	102.674153

Slide ID	SDMA/2020/Assam/18
Source	SDMA
Slide No	18
State	Assam
District	Cachar
Name	SDMA/AS/2020/SONAI5
Location	GOBINDANAGAR PT II
NH-SH Cutoff/Affected	
Lat (dd)	24.721872
Long (dd)	92.903858
Date of Event	04.06.2020
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	64.6480234

Slide ID	Newspaper&Media/2020/Assam/19
Source	Newspaper&Media
Slide No	19
State	Assam
District	Kamrup metro
Name	Jorabat Landslide
Location	12th mile area, Jorabat
NH-SH Cutoff/Affected	
Lat (dd)	26.102
Long (dd)	91.863
Date of Event	12.07.2020
Time of Event	
Type of Event	1
Casualty	2 dead
Rainfall (mm)	45.49959715

Slide ID	Newspaper&Media/2020/Assam/20
Source	Newspaper&Media
Slide No	20
State	Assam
District	Kamrup metro
Name	Guwahati Landslide
Location	Kharguli, Guwahati
NH-SH Cutoff/Affected	
Lat (dd)	26.196
Long (dd)	91.765
Date of Event	28.06.2020
Time of Event	
Type of Event	2
Casualty	1 dead
Rainfall (mm)	102.7784428

Slide ID	Newspaper&Media/2020/Manipur/26
Source	Newspaper&Media
Slide No	26
State	Manipur
District	Tamenglong
Name	Imphal-Jiribam Landslide-1
Location	NH cutoff at Oinamlong
NH-SH Cutoff/Affected	NH 37
Lat (dd)	24.774
Long (dd)	93.284
Date of Event	2-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	89.27880505

Slide ID	Newspaper&Media/2020/Manipur/27
Source	Newspaper&Media
Slide No	27
State	Manipur
District	Tamenglong
Name	Imphal-Jiribam Landslide-1
Location	NH cutoff at Khangsang
NH-SH Cutoff/Affected	NH 37
Lat (dd)	24.8205
Long (dd)	93.473
Date of Event	2-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	50.60031752

Slide ID	Newspaper&Media/2020/Manipur/28
Source	Newspaper&Media
Slide No	28
State	Manipur
District	Tamenglong
Name	
Location	Makru Bridge
NH-SH Cutoff/Affected	NH 37
Lat (dd)	24.8172
Long (dd)	93.2457
Date of Event	
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	

Slide ID	Newspaper&Media/2020/Manipur/29
Source	Newspaper&Media
Slide No	29
State	Manipur
District	Senapati
Name	Imphal-Jiribam Landslide-1
Location	Barak Bridge
NH-SH Cutoff/Affected	NH 37
Lat (dd)	25.396
Long (dd)	94.142
Date of Event	2-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	5.95963219

Slide ID	Newspaper&Media/2020/Manipur/30
Source	Newspaper&Media
Slide No	30
State	Manipur
District	Senapati
Name	Twilang Landslide
Location	Twilang area between 10 miles Poudel Basti and 13 miles Thonglang Atongba which is about 20 km away from Kangopoki DHQ
NH-SH Cutoff/Affected	Imphal-Tamenglong Road
Lat (dd)	25.236
Long (dd)	93.911
Date of Event	6-Jul-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	9.543156357

Slide ID	Newspaper&Media/2020/Manipur/31
Source	Newspaper&Media
Slide No	31
State	Manipur
District	Imphal east
Name	Khenjonglang Landslide
Location	Awa Ching located at Keirao Khunao
NH-SH Cutoff/Affected	
Lat (dd)	24.748
Long (dd)	93.986
Date of Event	11-Aug-20
Time of Event	4:00 A.M
Type of Event	1
Casualty	
Rainfall (mm)	18.95101404

Slide ID	Newspaper&Media/2020/Manipur/32
Source	Newspaper&Media
Slide No	32
State	Manipur
District	Senapati
Name	K Sinam Village_NH 37
Location	K Sinam village
NH-SH Cutoff/Affected	NH 37
Lat (dd)	24.925
Long (dd)	93.985
Date of Event	21-Jun-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	10.16939571

Slide ID	Newspaper&Media/2020/Manipur/33
Source	Newspaper&Media
Slide No	33
State	Manipur
District	Senapati
Name	Laii Village
Location	Lai Village, Paotama
NH-SH Cutoff/Affected	
Lat (dd)	25.498
Long (dd)	94.387
Date of Event	20-Aug-20
Time of Event	02:45 A.M
Type of Event	2
Casualty	
Rainfall (mm)	2.32650696

Slide ID	Newspaper&Media/2020/Manipur/34
Source	Newspaper&Media
Slide No	34
State	Manipur
District	Senapati
Name	Kotlen
Location	Kotlen
NH-SH Cutoff/Affected	NH 37
Lat (dd)	24.792
Long (dd)	93.696
Date of Event	28-Aug-20
Time of Event	11: 00 A.M
Type of Event	2
Casualty	
Rainfall (mm)	

Slide ID	Newspaper&Media/2020/Manipur/35
Source	Newspaper&Media
Slide No	35
State	Manipur
District	Chandel
Name	Khongkhang
Location	Khongkhang, Tengnoupal
NH-SH Cutoff/Affected	Imphal-Moreh Road
Lat (dd)	24.318
Long (dd)	94.222
Date of Event	20-Aug-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	65.30426639

Slide ID	Newspaper&Media/2020/Manipur/36
Source	Newspaper&Media
Slide No	36
State	Manipur
District	Tamenglong
Name	Imphal-Jiribam Landslide-2
Location	5 km from Nungba Bazaar towards Imphal
NH-SH Cutoff/Affected	NH 2
Lat (dd)	24.775278
Long (dd)	93.438889
Date of Event	27-May-20
Time of Event	3: 00 A.M
Type of Event	3
Casualty	
Rainfall (mm)	62.67473653

Slide ID	Newspaper&Media/2020/Manipur/37
Source	Newspaper&Media
Slide No	37
State	Manipur
District	Senapati
Name	Khongnem-Thana
Location	Khongnem Thana
NH-SH Cutoff/Affected	NH 2 - Imphal-Mao Sector
Lat (dd)	25.355956
Long (dd)	94.065278
Date of Event	23-Oct-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	60.30634994

Slide ID	GSI_REPORTS/2020/Manipur/38
Source	GSI_REPORTS
Slide No	38
State	Manipur
District	Tamenglong
Name	Longmai Slide
Location	Longmai Khumji Noney Area
NH-SH Cutoff/Affected	NH -37, Imphal - Jiribham Road, Longmai
Lat (dd)	24.855361
Long (dd)	93.624028
Date of Event	Last week of June
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	0.08182246

Slide ID	GSI_REPORTS/2020/Manipur/39
Source	GSI_REPORTS
Slide No	39
State	Manipur
District	Tamenglong
Name	Khumji Noney Slide
Location	100 meters from Noney bazaar towards Jiribam on the NH-37, Imphal-Jiribam road.
NH-SH Cutoff/Affected	
Lat (dd)	24.862694
Long (dd)	93.624056
Date of Event	Last week of June
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	0.08182246

Slide ID	GSI_REPORTS/2020/Manipur/40
Source	GSI_REPORTS
Slide No	40
State	Manipur
District	Senapati
Name	Mao Slide
Location	Song Song Village Mao Gate
NH-SH Cutoff/Affected	
Lat (dd)	25.515917
Long (dd)	94.136528
Date of Event	Sometime before 21 July
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	6.404231947

MEGHALAYA

Slide ID	GSI_REPORTS/2020/Meghalaya/41
Source	GSI_REPORTS
Slide No	41
State	Meghalaya
District	East Khasi Hills
Name	Umjapung slide
Location	Mawiong Umjapung
NH-SH Cutoff/Affected	
Lat (dd)	25.62
Long (dd)	91.882722
Date of Event	21-May-20
Time of Event	1:00 P.M
Type of Event	1
Casualty	
Rainfall (mm)	60.05292242

Slide ID	GSI_REPORTS/2020/Meghalaya/42
Source	GSI_REPORTS
Slide No	42
State	Meghalaya
District	Jaintia Hills
Name	lawmusiang Landslide - I
Location	lawmusiang Wahthanad
NH-SH Cutoff/Affected	
Lat (dd)	25.442333
Long (dd)	92.194472
Date of Event	26-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	47.53227393

Slide ID	GSI_REPORTS/2020/Meghalaya/43
Source	GSI_REPORTS
Slide No	43
State	Meghalaya
District	Jaintia Hills
Name	lawmusiang Landslide – II
Location	lawmusiang Wahthanad
NH-SH Cutoff/Affected	
Lat (dd)	25.441778
Long (dd)	92.195583
Date of Event	26-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	47.53227393

Slide ID	GSI_REPORTS/2020/Meghalaya/44
Source	GSI_REPORTS
Slide No	44
State	Meghalaya
District	Jaintia Hills
Name	Riatsiatsim Landslide
Location	Riatsiatsim
NH-SH Cutoff/Affected	
Lat (dd)	25.441389
Long (dd)	92.191083
Date of Event	26-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	47.53227393

Slide ID	GSI_REPORTS/2020/Meghalaya/45
Source	GSI_REPORTS
Slide No	45
State	Meghalaya
District	Jaintia Hills
Name	Chutwakhu road landslide
Location	Chutwakhu Rd near Govt. Boy's Higher secondary school
NH-SH Cutoff/Affected	
Lat (dd)	25.436306
Long (dd)	92.19525
Date of Event	26-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	47.53227393

Slide ID	GSI_REPORTS/2020/Meghalaya/46
Source	GSI_REPORTS
Slide No	46
State	Meghalaya
District	Jaintia Hills
Name	Chiliangraij Landslide
Location	Chiliangraij
NH-SH Cutoff/Affected	
Lat (dd)	25.435056
Long (dd)	92.198361
Date of Event	26-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	47.53227393

Slide ID	GSI_REPORTS/2020/Meghalaya/47
Source	GSI_REPORTS
Slide No	47
State	Meghalaya
District	Jaintia Hills
Name	Salaroh Landslide
Location	Salaroh
NH-SH Cutoff/Affected	
Lat (dd)	25.447417
Long (dd)	92.201389
Date of Event	26-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	54.20739815

Slide ID	GSI_REPORTS/2020/Meghalaya/48
Source	GSI_REPORTS
Slide No	48
State	Meghalaya
District	Jaintia Hills
Name	Caroline-Jowai
Location	Caroline colony
NH-SH Cutoff/Affected	
Lat (dd)	25.45075
Long (dd)	92.194528
Date of Event	26-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	47.53227393

Slide ID	GSI_REPORTS/2020/Meghalaya/49
Source	GSI_REPORTS
Slide No	49
State	Meghalaya
District	Jaintia Hills
Name	Shillong-Agartala-Sabrum Rd
Location	Shillong-Agartala-Sabrum Rd
NH-SH Cutoff/Affected	
Lat (dd)	25.453517
Long (dd)	92.187847
Date of Event	26-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	47.53227393

Slide ID	Newspaper&Media/2020/Meghalaya/50
Source	Newspaper&Media
Slide No	50
State	Meghalaya
District	East Khasi Hills
Name	Laitlyngkot-Pomlum
Location	between Laitlyngkot and Pomlum
NH-SH Cutoff/Affected	
Lat (dd)	25.448
Long (dd)	91.84
Date of Event	11-Jul-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	473.2768601

Slide ID	FIELD_VISIT/2020/Meghalaya/51
Source	FIELD_VISIT
Slide No	51
State	Meghalaya
District	East Khasi Hills
Name	Thangshalai
Location	Thangshalai, Diengpasoh
NH-SH Cutoff/Affected	
Lat (dd)	25.590667
Long (dd)	92.059889
Date of Event	27-Sep-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	26.57811355

Slide ID	Newspaper&Media/2020/Meghalaya/52
Source	Newspaper&Media
Slide No	52
State	Meghalaya
District	Ri-bhoi
Name	Shobhala
Location	NH 06, Umsning
NH-SH Cutoff/Affected	NH 06
Lat (dd)	25.746
Long (dd)	91.889
Date of Event	24-Oct-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	229.6170364

Slide ID	Newspaper&Media/2020/Meghalaya/53
Source	Newspaper&Media
Slide No	53
State	Meghalaya
District	West Khasi Hills
Name	Mawthawiang
Location	Jashiar-Nongshilong road
NH-SH Cutoff/Affected	Jashiar-Nongshilong Road
Lat (dd)	25.43
Long (dd)	91.471
Date of Event	10-Jul-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	591.3999885

Slide ID	Newspaper&Media/2020/Meghalaya/54
Source	Newspaper&Media
Slide No	54
State	Meghalaya
District	West Khasi Hills
Name	Mawthawpdah-Rangdikhew road
Location	Mawthawpdah-Rangdikhew road
NH-SH Cutoff/Affected	Mawthawpdah-Rangdikhew road
Lat (dd)	25.417
Long (dd)	91.416
Date of Event	9-Jul-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	94.6425512

Slide ID	Newspaper&Media/2020/Meghalaya/55
Source	Newspaper&Media
Slide No	55
State	Meghalaya
District	West Khasi Hills
Name	Nongstoin – Tura By-pass
Location	PHE pipelines which connect to Pyndengrei village
NH-SH Cutoff/Affected	
Lat (dd)	25.528
Long (dd)	91.262
Date of Event	11-Jul-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	200.1913643

Slide ID	Newspaper&Media/2020/Meghalaya/56
Source	Newspaper&Media
Slide No	56
State	Meghalaya
District	East Khasi Hills
Name	Police Reserve
Location	Police Reserve Shillong
NH-SH Cutoff/Affected	
Lat (dd)	25.581
Long (dd)	91.884
Date of Event	12-Jul-20
Time of Event	
Type of Event	2
Casualty	2 dead
Rainfall (mm)	160.51772

Slide ID	Newspaper&Media/2020/Meghalaya/57
Source	Newspaper&Media
Slide No	57
State	Meghalaya
District	East Khasi Hills
Name	Dhobi Ghat Landslide
Location	Parmaw Dhobi Ghat Laban
NH-SH Cutoff/Affected	
Lat (dd)	25.561
Long (dd)	91.88
Date of Event	25-Sep-20
Time of Event	Early morning
Type of Event	2
Casualty	5 dead
Rainfall (mm)	277.9193161

Slide ID	Newspaper&Media/2020/Meghalaya/58
Source	Newspaper&Media
Slide No	58
State	Meghalaya
District	West Garo Hills
Name	Tura Highway landslide
Location	highway at Dollonggre, connecting Tura with Paikan,
NH-SH Cutoff/Affected	
Lat (dd)	25.726
Long (dd)	90.394
Date of Event	24-Jul-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	8.911403826

Slide ID	Newspaper&Media/2020/Meghalaya/59
Source	Newspaper&Media
Slide No	59
State	Meghalaya
District	West Khasi Hills
Name	Nongstoin Block C&RD
Location	landslides occurred in six villages under Nongstoin, Mawthadraishan and Mawshynrut C&RD blocks
NH-SH Cutoff/Affected	
Lat (dd)	25.523
Long (dd)	91.272
Date of Event	14-Jul-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	2.044385709

Slide ID	Newspaper&Media/2020/Meghalaya/60
Source	Newspaper&Media
Slide No	60
State	Meghalaya
District	West Khasi Hills
Name	Mawthadraishan C &RD
Location	landslides occurred in six villages under Nongstoin, Mawthadraishan and Mawshynrut C&RD blocks
NH-SH Cutoff/Affected	
Lat (dd)	25.516
Long (dd)	91.469
Date of Event	14-Jul-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	2.358563167

Slide ID	Newspaper&Media/2020/Meghalaya/61
Source	Newspaper&Media
Slide No	61
State	Meghalaya
District	West Khasi Hills
Name	Mawshynrut C & D block
Location	landslides occurred in six villages under Nongstoin, Mawthadraishan and Mawshynrut C&RD blocks
NH-SH Cutoff/Affected	
Lat (dd)	25.676
Long (dd)	91.055
Date of Event	14-Jul-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	3.429939509

Slide ID	Newspaper&Media/2020/Meghalaya/62
Source	Newspaper&Media
Slide No	62
State	Meghalaya
District	East Khasi Hills
Name	landslide at Ryndang briew
Location	Ryndang briew, Umiam
NH-SH Cutoff/Affected	
Lat (dd)	25.65
Long (dd)	91.893
Date of Event	24-Sep-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	124.5995509

Slide ID	Newspaper&Media/2020/Meghalaya/63
Source	Newspaper&Media
Slide No	63
State	Meghalaya
District	East Khasi Hills
Name	18th km of Mawkdok Khadarshnong Road
Location	Mawkdok Khadarshnong Road
NH-SH Cutoff/Affected	Mawkdok Khadarshnong Road
Lat (dd)	25.4
Long (dd)	91.786
Date of Event	29-May-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	

MIZORAM

Slide ID	SDMA/2020/Mizoram/64
Source	SDMA
Slide No	64
State	Mizoram
District	Aizawl
Name	Zembawk_1
Location	Zemabawk
NH-SH Cutoff/Affected	
Lat (dd)	23.731928
Long (dd)	92.760028
Date of Event	06.10.2020
Time of Event	3:05 P.M
Type of Event	1
Casualty	
Rainfall (mm)	84.12998903

Slide ID	SDMA/2020/Mizoram/65
Source	SDMA
Slide No	65
State	Mizoram
District	Aizawl
Name	Bawngkawn_1
Location	Bawngkawn
NH-SH Cutoff/Affected	
Lat (dd)	23.721556
Long (dd)	92.731186
Date of Event	22.09.2020
Time of Event	3:00 A.M
Type of Event	1
Casualty	
Rainfall (mm)	10.2034105

Slide ID	SDMA/2020/Mizoram/66
Source	SDMA
Slide No	66
State	Mizoram
District	Aizawl
Name	Venglai_1
Location	Venglai
NH-SH Cutoff/Affected	
Lat (dd)	23.740453
Long (dd)	92.721578
Date of Event	01.10.2020
Time of Event	8:30 P.M
Type of Event	1
Casualty	
Rainfall (mm)	28.04460382

Slide ID	SDMA/2020/Mizoram/67
Source	SDMA
Slide No	67
State	Mizoram
District	Aizawl
Name	Maubawk_1
Location	Maubawk
NH-SH Cutoff/Affected	
Lat (dd)	23.720156
Long (dd)	92.698094
Date of Event	05.06.2020
Time of Event	11:00 A.M
Type of Event	1
Casualty	
Rainfall (mm)	43.24052771

Slide ID	SDMA/2020/Mizoram/68
Source	SDMA
Slide No	68
State	Mizoram
District	Aizawl
Name	Maubawk_2
Location	Maubawk
NH-SH Cutoff/Affected	
Lat (dd)	23.723239
Long (dd)	92.700333
Date of Event	27.09.2020
Time of Event	3:05 P.M
Type of Event	1
Casualty	
Rainfall (mm)	13.21906561

Slide ID	SDMA/2020/Mizoram/69
Source	SDMA
Slide No	69
State	Mizoram
District	Aizawl
Name	Maubawk_3
Location	Maubawk
NH-SH Cutoff/Affected	
Lat (dd)	23.751072
Long (dd)	92.695947
Date of Event	13.08.2020
Time of Event	6:00 P.M
Type of Event	1
Casualty	
Rainfall (mm)	0.099278871

Slide ID	SDMA/2020/Mizoram/70
Source	SDMA
Slide No	70
State	Mizoram
District	
Name	Chanpui_1
Location	Chawnpui
NH-SH Cutoff/Affected	
Lat (dd)	23.732306
Long (dd)	94.706492
Date of Event	04.10.2020
Time of Event	6:00 AM
Type of Event	1
Casualty	
Rainfall (mm)	17.14106971

Slide ID	SDMA/2020/Mizoram/71
Source	SDMA
Slide No	71
State	Mizoram
District	Aizawl
Name	Vaivakawn_1
Location	Vaivakawn
NH-SH Cutoff/Affected	
Lat (dd)	23.739519
Long (dd)	92.71415
Date of Event	02.10.2020
Time of Event	9:00 P.M
Type of Event	1
Casualty	
Rainfall (mm)	32.33645279

Slide ID	SDMA/2020/Mizoram/72
Source	SDMA
Slide No	72
State	Mizoram
District	Aizawl
Name	Bawngkawn_2
Location	Bawngkawn
NH-SH Cutoff/Affected	
Lat (dd)	23.720222
Long (dd)	92.728675
Date of Event	04.07.2020
Time of Event	3:00 A.M
Type of Event	1
Casualty	
Rainfall (mm)	6.16730317

Slide ID	SDMA/2020/Mizoram/73
Source	SDMA
Slide No	73
State	Mizoram
District	Aizawl
Name	Maubuang_1
Location	Maubuang
NH-SH Cutoff/Affected	
Lat (dd)	23.488881
Long (dd)	92.708261
Date of Event	31.08.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	0.560721529

Slide ID	SDMA/2020/Mizoram/74
Source	SDMA
Slide No	74
State	Mizoram
District	Aizawl
Name	Khatla_1
Location	Khatla
NH-SH Cutoff/Affected	
Lat (dd)	23.720278
Long (dd)	92.716983
Date of Event	10.06.2020
Time of Event	Day
Type of Event	1
Casualty	
Rainfall (mm)	62.90297881

Slide ID	SDMA/2020/Mizoram/75
Source	SDMA
Slide No	75
State	Mizoram
District	Aizawl
Name	Melriat_1
Location	Melriat
NH-SH Cutoff/Affected	
Lat (dd)	23.562675
Long (dd)	92.726283
Date of Event	11.07.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	22.14058243

Slide ID	SDMA/2020/Mizoram/76
Source	SDMA
Slide No	76
State	Mizoram
District	Aizawl
Name	Zemabawk_2
Location	Zemabawk
NH-SH Cutoff/Affected	
Lat (dd)	23.730278
Long (dd)	92.757661
Date of Event	02.02.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	

Slide ID	SDMA/2020/Mizoram/77
Source	SDMA
Slide No	77
State	Mizoram
District	Aizawl
Name	Ramhlun_1
Location	Ramhlun
NH-SH Cutoff/Affected	
Lat (dd)	23.746222
Long (dd)	92.723003
Date of Event	15.06.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	33.24520684

Slide ID	SDMA/2020/Mizoram/78
Source	SDMA
Slide No	78
State	Mizoram
District	Aizawl
Name	Vaivakawn_2
Location	Vaivakawn
NH-SH Cutoff/Affected	
Lat (dd)	23.739747
Long (dd)	92.708728
Date of Event	17.07.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	2.030592573

Slide ID	SDMA/2020/Mizoram/79
Source	SDMA
Slide No	79
State	Mizoram
District	Aizawl
Name	Mualpheng_1
Location	Mualpheng
NH-SH Cutoff/Affected	
Lat (dd)	23.612047
Long (dd)	92.974247
Date of Event	28.05.2020
Time of Event	Day
Type of Event	1
Casualty	
Rainfall (mm)	183.3699088

Slide ID	SDMA/2020/Mizoram/80
Source	SDMA
Slide No	80
State	Mizoram
District	Aizawl
Name	Sailam_1
Location	Sailam
NH-SH Cutoff/Affected	
Lat (dd)	23.350964
Long (dd)	92.797306
Date of Event	10.07.2020
Time of Event	11:00 A.M
Type of Event	1
Casualty	
Rainfall (mm)	15.71465027

Slide ID	SDMA/2020/Mizoram/81
Source	SDMA
Slide No	81
State	Mizoram
District	Champhai
Name	Hnahlan-1
Location	Hnahlan
NH-SH Cutoff/Affected	
Lat (dd)	23.692778
Long (dd)	93.382778
Date of Event	08.08.2020
Time of Event	Evening
Type of Event	1
Casualty	
Rainfall (mm)	18.256644

Slide ID	SDMA/2020/Mizoram/82
Source	SDMA
Slide No	82
State	Mizoram
District	Champhai
Name	Zotlang-1
Location	Zotlang
NH-SH Cutoff/Affected	
Lat (dd)	23.692778
Long (dd)	93.382778
Date of Event	01.08.2020
Time of Event	Evening
Type of Event	1
Casualty	
Rainfall (mm)	31.91210837

Slide ID	SDMA/2020/Mizoram/83
Source	SDMA
Slide No	83
State	Mizoram
District	Lawngtlai
Name	Electric_veng1
Location	Electric veng
NH-SH Cutoff/Affected	
Lat (dd)	22.536667
Long (dd)	92.887222
Date of Event	07.07.2020
Time of Event	Evening
Type of Event	1
Casualty	
Rainfall (mm)	0.308505963

Slide ID	SDMA/2020/Mizoram/84
Source	SDMA
Slide No	84
State	Mizoram
District	Lawngtlai
Name	Electric_veng2
Location	Electric veng
NH-SH Cutoff/Affected	
Lat (dd)	22.536111
Long (dd)	92.889722
Date of Event	03.08.2020
Time of Event	Evening
Type of Event	1
Casualty	
Rainfall (mm)	10.63611803

Slide ID	SDMA/2020/Mizoram/85
Source	SDMA
Slide No	85
State	Mizoram
District	Serchhip
Name	Electric_veng3
Location	Electric veng
NH-SH Cutoff/Affected	
Lat (dd)	23.533611
Long (dd)	92.891944
Date of Event	04.08.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	18.01209834

Slide ID	SDMA/2020/Mizoram/86
Source	SDMA
Slide No	86
State	Mizoram
District	Lawngtlai
Name	College_veng1
Location	College Veng
NH-SH Cutoff/Affected	
Lat (dd)	22.533611
Long (dd)	92.5575
Date of Event	03.08.2020
Time of Event	Evening
Type of Event	1
Casualty	
Rainfall (mm)	7.022304045

Slide ID	SDMA/2020/Mizoram/87
Source	SDMA
Slide No	87
State	Mizoram
District	Lawngtlai
Name	Electric_veng4
Location	Electric veng
NH-SH Cutoff/Affected	
Lat (dd)	22.534167
Long (dd)	92.889444
Date of Event	20.08.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	20.45519679

Slide ID	SDMA/2020/Mizoram/88
Source	SDMA
Slide No	88
State	Mizoram
District	Serchhip
Name	Electric_veng5
Location	Electric veng
NH-SH Cutoff/Affected	
Lat (dd)	23.533611
Long (dd)	92.890833
Date of Event	24.08.2020
Time of Event	Evening
Type of Event	1
Casualty	
Rainfall (mm)	17.69289468

Slide ID	SDMA/2020/Mizoram/89
Source	SDMA
Slide No	89
State	Mizoram
District	Lawngtlai
Name	College_veng2
Location	College Veng
NH-SH Cutoff/Affected	
Lat (dd)	22.515556
Long (dd)	92.896389
Date of Event	02.08.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	32.07670913

Slide ID	SDMA/2020/Mizoram/90
Source	SDMA
Slide No	90
State	Mizoram
District	Lawngtlai
Name	Bazaar_veng1
Location	Bazar veng
NH-SH Cutoff/Affected	
Lat (dd)	22.527222
Long (dd)	92.898889
Date of Event	03.08.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	10.63611803

Slide ID	SDMA/2020/Mizoram/91
Source	SDMA
Slide No	91
State	Mizoram
District	Aizawl
Name	Rulchawm1
Location	Rulchawm
NH-SH Cutoff/Affected	
Lat (dd)	23.6625
Long (dd)	92.9675
Date of Event	02.10.2020
Time of Event	7:45 P.M
Type of Event	1
Casualty	
Rainfall (mm)	33.03127081

Slide ID	SDMA/2020/Mizoram/92
Source	SDMA
Slide No	92
State	Mizoram
District	Aizawl
Name	Phullen1
Location	Phullen
NH-SH Cutoff/Affected	
Lat (dd)	23.85
Long (dd)	93.0475
Date of Event	02.10.2020 – 03.10.2020
Time of Event	8:00 P.M
Type of Event	1
Casualty	
Rainfall (mm)	88.50882052

Slide ID	SDMA/2020/Mizoram/93
Source	SDMA
Slide No	93
State	Mizoram
District	Aizawl
Name	Phullen2
Location	Phullen
NH-SH Cutoff/Affected	
Lat (dd)	23.855617
Long (dd)	93.0475
Date of Event	17.08.2020
Time of Event	07:45 P.M
Type of Event	1
Casualty	
Rainfall (mm)	0.139587213

Slide ID	SDMA/2020/Mizoram/94
Source	SDMA
Slide No	94
State	Mizoram
District	Aizawl
Name	Phullen3
Location	Phullen
NH-SH Cutoff/Affected	
Lat (dd)	23.841111
Long (dd)	93.051944
Date of Event	13.10.2020
Time of Event	4:30 P.M
Type of Event	1
Casualty	
Rainfall (mm)	0.201731496

Slide ID	SDMA/2020/Mizoram/95
Source	SDMA
Slide No	95
State	Mizoram
District	Dhalai
Name	Phullen4
Location	Phullen
NH-SH Cutoff/Affected	
Lat (dd)	23.838611
Long (dd)	92.053611
Date of Event	08.10.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	0.583468754

Slide ID	SDMA/2020/Mizoram/96
Source	SDMA
Slide No	96
State	Mizoram
District	Aizawl
Name	Phullen5
Location	Phullen
NH-SH Cutoff/Affected	
Lat (dd)	23.673333
Long (dd)	92.885556
Date of Event	10.07.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	6.874414171

Slide ID	SDMA/2020/Mizoram/97
Source	SDMA
Slide No	97
State	Mizoram
District	Mamit
Name	Mamit1
Location	Mamit
NH-SH Cutoff/Affected	
Lat (dd)	23.92
Long (dd)	92.491111
Date of Event	10.06.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	49.12471811

Slide ID	SDMA/2020/Mizoram/98
Source	SDMA
Slide No	98
State	Mizoram
District	Mamit
Name	Mamit2
Location	Mamit
NH-SH Cutoff/Affected	
Lat (dd)	23.916944
Long (dd)	92.492222
Date of Event	11.07.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	14.19579004

Slide ID	SDMA/2020/Mizoram/99
Source	SDMA
Slide No	99
State	Mizoram
District	Mamit
Name	Mamit3
Location	Mamit
NH-SH Cutoff/Affected	
Lat (dd)	23.924167
Long (dd)	92.490278
Date of Event	11.06.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	76.22800361

Slide ID	SDMA/2020/Mizoram/100
Source	SDMA
Slide No	100
State	Mizoram
District	Mamit
Name	Mamit4
Location	Mamit
NH-SH Cutoff/Affected	
Lat (dd)	23.912778
Long (dd)	92.390556
Date of Event	11.06.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	118.1136405

Slide ID	SDMA/2020/Mizoram/101
Source	SDMA
Slide No	101
State	Mizoram
District	Mamit
Name	Mamit5
Location	Mamit
NH-SH Cutoff/Affected	
Lat (dd)	23.948611
Long (dd)	92.49
Date of Event	29.05.2020
Time of Event	Day
Type of Event	1
Casualty	
Rainfall (mm)	0.101542908

Slide ID	SDMA/2020/Mizoram/102
Source	SDMA
Slide No	102
State	Mizoram
District	Champhai
Name	Khazawl1
Location	Khazawl
NH-SH Cutoff/Affected	
Lat (dd)	23.523083
Long (dd)	93.193194
Date of Event	03.10.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	67.90231376

Slide ID	SDMA/2020/Mizoram/103
Source	SDMA
Slide No	103
State	Mizoram
District	Kolasib
Name	Tumpui_veng1
Location	Tumpui Veng
NH-SH Cutoff/Affected	
Lat (dd)	24.231944
Long (dd)	92.675278
Date of Event	21.10.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	22.9018619

Slide ID	SDMA/2020/Mizoram/104
Source	SDMA
Slide No	104
State	Mizoram
District	Kolasib
Name	Variengte_IOC_veng1
Location	Variengte IOC Veng
NH-SH Cutoff/Affected	
Lat (dd)	24.455
Long (dd)	92.741667
Date of Event	25.05.2020
Time of Event	Night
Type of Event	1
Casualty	
Rainfall (mm)	2.635972523

Slide ID	Newspaper&Media/2020/Nagaland/105
Source	Newspaper&Media
Slide No	105
State	Nagaland
District	Kohima
Name	Lower PWD
Location	Lower PWD, Kohima
NH-SH Cutoff/Affected	
Lat (dd)	25.662
Long (dd)	94.106
Date of Event	4-Oct-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	7.506611078

Slide ID	Newspaper&Media/2020/Nagaland/106
Source	Newspaper&Media
Slide No	106
State	Nagaland
District	Mokokchung
Name	Alongmen Ward, Mokokchung
Location	Mokokchung District
NH-SH Cutoff/Affected	
Lat (dd)	26.326
Long (dd)	94.522
Date of Event	15-Aug-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	118.9269139

Slide ID	Newspaper&Media/2020/Nagaland/107
Source	Newspaper&Media
Slide No	107
State	Nagaland
District	Zunheboto
Name	DC Hill West Colony
Location	DC Hill West Colony, Zunheboto District
NH-SH Cutoff/Affected	
Lat (dd)	26.01
Long (dd)	94.519
Date of Event	17-Aug-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	15.68829081

Slide ID	Newspaper&Media/2020/Nagaland/108
Source	Newspaper&Media
Slide No	108
State	Nagaland
District	Zunheboto
Name	Akuluto Town
Location	Zunheboto District
NH-SH Cutoff/Affected	
Lat (dd)	26.166
Long (dd)	94.487
Date of Event	14-Aug-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	11.64369563

Slide ID	Newspaper&Media/2020/Nagaland/109
Source	Newspaper&Media
Slide No	109
State	Nagaland
District	Dimapur
Name	Patkai Bridge to Kukidolong road stretch
Location	Near Nagaland Welcome Gate
NH-SH Cutoff/Affected	NH 29
Lat (dd)	25.79641717
Long (dd)	93.8023278
Date of Event	7-Sep-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	10.12091863

Slide ID	Newspaper&Media/2020/Nagaland/110
Source	Newspaper&Media
Slide No	110
State	Nagaland
District	Dimapur
Name	Patkai Bridge to Kukidolong road stretch
Location	Dimapur-Kohima 4 lane highway
NH-SH Cutoff/Affected	NH 29
Lat (dd)	25.79383239
Long (dd)	93.80556258
Date of Event	12-Sep-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	7.490973544

Slide ID	Newspaper&Media/2020/Nagaland/111
Source	Newspaper&Media
Slide No	111
State	Nagaland
District	Dimapur
Name	Patkai Bridge to Kukidolong road strech
Location	Kukidalong, 3 km from Chumokedima Check Gate
NH-SH Cutoff/Affected	NH 29
Lat (dd)	25.77079334
Long (dd)	93.81473269
Date of Event	19-Sep-20
Time of Event	5:00 A.M.
Type of Event	3
Casualty	
Rainfall (mm)	1.495132544

Slide ID	Newspaper&Media/2020/Nagaland/112
Source	Newspaper&Media
Slide No	112
State	Nagaland
District	Kohima
Name	Mehruliasta, Paramedical Colony
Location	Timber mill, Pramedical colony, Kohima District
NH-SH Cutoff/Affected	
Lat (dd)	25.664
Long (dd)	94.091
Date of Event	19-Aug-20
Time of Event	11:45 A.M
Type of Event	3
Casualty	
Rainfall (mm)	4.507467725

Slide ID	Newspaper&Media/2020/Nagaland/113
Source	Newspaper&Media
Slide No	113
State	Nagaland
District	Zunheboto
Name	Between Xuivi and Kilomi District
Location	40 km from Zunheboto Town between Xuivi and Kilomi
NH-SH Cutoff/Affected	
Lat (dd)	25.873
Long (dd)	94.441
Date of Event	8-Jul-20
Time of Event	
Type of Event	3
Casualty	1 dead
Rainfall (mm)	11.88718098

Slide ID	Newspaper&Media/2020/Nagaland/114
Source	Newspaper&Media
Slide No	114
State	Nagaland
District	Kohima
Name	High School Area
Location	High School Area, Kohima
NH-SH Cutoff/Affected	
Lat (dd)	25.689
Long (dd)	94.098
Date of Event	11-Jul-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	29.49197781

Slide ID	Newspaper&Media/2020/Sikkim/115
Source	Newspaper&Media
Slide No	115
State	Sikkim
District	West Sikkim
Name	Lingchom Landslide
Location	Kabi Lungstog Constituency
NH-SH Cutoff/Affected	
Lat (dd)	27.286
Long (dd)	88.219
Date of Event	24-May-20
Time of Event	12:30 P.M
Type of Event	2
Casualty	1 dead
Rainfall (mm)	

Slide ID	Newspaper&Media/2020/Sikkim/116
Source	Newspaper&Media
Slide No	116
State	Sikkim
District	East Sikkim
Name	Gangtok_Rangpo_landslides
Location	Between Gangtok and Rangpo on NH 10
NH-SH Cutoff/Affected	NH 10
Lat (dd)	27.261
Long (dd)	88.553
Date of Event	8-Sep-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	6.692192465

Slide ID	Newspaper&Media/2020/Sikkim/117
Source	Newspaper&Media
Slide No	117
State	Sikkim
District	East Sikkim
Name	20 Mile Bardun
Location	Between Gangtok and Rangpo on NH 10
NH-SH Cutoff/Affected	
Lat (dd)	27.286
Long (dd)	88.59
Date of Event	8-Sep-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	6.692192465

Slide ID	Newspaper&Media/2020/Sikkim/118
Source	Newspaper&Media
Slide No	118
State	Sikkim
District	East Sikkim
Name	Gai Dhara Landslide
Location	Between Gangtok and Rangpo on NH 10
NH-SH Cutoff/Affected	
Lat (dd)	27.256
Long (dd)	88.546
Date of Event	8-Sep-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	6.692192465

Slide ID	Newspaper&Media/2020/Sikkim/119
Source	Newspaper&Media
Slide No	119
State	Sikkim
District	South Sikkim
Name	Thakka Road mudslide
Location	Thakka Road, Ben , South Sikkim
NH-SH Cutoff/Affected	
Lat (dd)	27.257
Long (dd)	88.421
Date of Event	8-Jun-20
Time of Event	
Type of Event	2
Casualty	2 dead
Rainfall (mm)	

Slide ID	Newspaper&Media/2020/Sikkim/120
Source	Newspaper&Media
Slide No	120
State	Sikkim
District	West Sikkim
Name	Boom Tafel
Location	Boom Tafel under Samdong Sribadam GPU, Rinchenpong
NH-SH Cutoff/Affected	
Lat (dd)	27.2206
Long (dd)	88.2034
Date of Event	2-Oct-20
Time of Event	8:30 P.M.
Type of Event	2
Casualty	2 dead
Rainfall (mm)	0.156456914

Slide ID	Newspaper&Media/2020/Sikkim/121
Source	Newspaper&Media
Slide No	121
State	Sikkim
District	North Sikkim
Name	Mangang-Chungtohong Road
Location	
NH-SH Cutoff/Affected	Mangang-Chungtohong Road
Lat (dd)	27.502
Long (dd)	88.535
Date of Event	29-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	

Slide ID	Newspaper&Media/2020/Sikkim/122
Source	Newspaper&Media
Slide No	122
State	Sikkim
District	North Sikkim
Name	Barfok-Lingdok GPU
Location	
NH-SH Cutoff/Affected	
Lat (dd)	27.50519261
Long (dd)	88.50450975
Date of Event	27-Jun-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	24.42941415

Slide ID	Newspaper&Media/2020/Sikkim/123
Source	Newspaper&Media
Slide No	123
State	Sikkim
District	North Sikkim
Name	Hee Gyanthang
Location	
NH-SH Cutoff/Affected	
Lat (dd)	27.4569233
Long (dd)	88.48734731
Date of Event	27-Jun-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	3.398117594

Slide ID	Newspaper&Media/2020/Sikkim/124
Source	Newspaper&Media
Slide No	124
State	Sikkim
District	North Sikkim
Name	Lum Village
Location	
NH-SH Cutoff/Affected	
Lat (dd)	27.38950527
Long (dd)	88.47566224
Date of Event	27-Jun-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	1.822928986

Slide ID	Newspaper&Media/2020/Sikkim/125
Source	Newspaper&Media
Slide No	125
State	Sikkim
District	North Sikkim
Name	Uphill Panang Landslide
Location	Upper Panang, Passingdong Village
NH-SH Cutoff/Affected	
Lat (dd)	27.52
Long (dd)	88.509
Date of Event	27-Jun-20
Time of Event	8:00 P.M
Type of Event	2
Casualty	
Rainfall (mm)	24.42941415

Slide ID	Newspaper&Media/2020/Sikkim/126
Source	Newspaper&Media
Slide No	126
State	Sikkim
District	North Sikkim
Name	Mangan-Sangkhalang Road
Location	
NH-SH Cutoff/Affected	Mangan-Sangkhalang Road
Lat (dd)	27.506
Long (dd)	88.528
Date of Event	27-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	24.42941415

Slide ID	Newspaper&Media/2020/Sikkim/127
Source	Newspaper&Media
Slide No	127
State	Sikkim
District	West Sikkim
Name	Landslide near Fambrong Falls
Location	Pelling-Tashiding Road; Tashiding to Gerethang Road; Reshi-Rinchepong Road
NH-SH Cutoff/Affected	Pelling-Tashiding Road
Lat (dd)	27.357
Long (dd)	88.232
Date of Event	28-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	2.882167201

Slide ID	Newspaper&Media/2020/Sikkim/128
Source	Newspaper&Media
Slide No	128
State	Sikkim
District	East Sikkim
Name	Tessta V NHPC Landslide
Location	Tessta V dam, NHPC, Apdara East Sikkim
NH-SH Cutoff/Affected	
Lat (dd)	27.251
Long (dd)	88.459
Date of Event	27-Jun-20
Time of Event	
Type of Event	2
Casualty	
Rainfall (mm)	0.420184951

Slide ID	Newspaper&Media/2020/Sikkim/129
Source	Newspaper&Media
Slide No	129
State	Sikkim
District	North Sikkim
Name	Myong Kyong 12th Mile Silde
Location	near Bey in Upper Dzongu
NH-SH Cutoff/Affected	
Lat (dd)	27.565
Long (dd)	88.424
Date of Event	26-May-20
Time of Event	5:00 P.M
Type of Event	2
Casualty	2 dead
Rainfall (mm)	0.361460652

Slide ID	Newspaper&Media/2020/Sikkim/130
Source	Newspaper&Media
Slide No	130
State	Sikkim
District	North Sikkim
Name	North_Sikkim_cutoff_event_1
Location	Ambithang near Rangrang and Ramthang under Namok Panchayat District
NH-SH Cutoff/Affected	NH10
Lat (dd)	27.411
Long (dd)	88.546
Date of Event	25-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	14.50819735

Slide ID	Newspaper&Media/2020/Sikkim/131
Source	Newspaper&Media
Slide No	131
State	Sikkim
District	North Sikkim
Name	North_Sikkim_cutoff_event_2
Location	Phamtham at Kabi Lungstok connecting North Sikkim to Tashi View Point
NH-SH Cutoff/Affected	NH10
Lat (dd)	27.453
Long (dd)	88.61
Date of Event	25-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	22.01125104

Slide ID	Newspaper&Media/2020/Sikkim/132
Source	Newspaper&Media
Slide No	132
State	Sikkim
District	East Sikkim
Name	North_Sikkim_cutoff_event_3
Location	Dikchu-Mangan Road at Tintek and Dikchu Sokpay
NH-SH Cutoff/Affected	NH10
Lat (dd)	27.384
Long (dd)	88.505
Date of Event	25-Jun-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	17.36318982

Slide ID	GSI_REPORTS/2020/Sikkim/133
Source	GSI_REPORTS
Slide No	133
State	Sikkim
District	North Sikkim
Name	Dhare Landslide
Location	Kabi District
NH-SH Cutoff/Affected	
Lat (dd)	27.398667
Long (dd)	88.623583
Date of Event	24-May-20
Time of Event	
Type of Event	1
Casualty	
Rainfall (mm)	

Slide ID	GSI_REPORTS/2020/Sikkim/134
Source	GSI_REPORTS
Slide No	134
State	Sikkim
District	North Sikkim
Name	Satala Landslide
Location	Upper Linchom under Kabi-Lungchuk District
NH-SH Cutoff/Affected	
Lat (dd)	27.398389
Long (dd)	88.625667
Date of Event	24-May-20
Time of Event	1:00 P.M
Type of Event	1
Casualty	
Rainfall (mm)	

Slide ID	Newspaper&Media/2020/Sikkim/135
Source	Newspaper&Media
Slide No	135
State	Sikkim
District	South Sikkim
Name	Sokpay_landslide
Location	The Dikchu Gangtok road blocked at Sopkay, Lower Rangdong about 3 kms short of Dikchu
NH-SH Cutoff/Affected	NH10
Lat (dd)	27.39
Long (dd)	88.429
Date of Event	1-Jun-20
Time of Event	7:00 P.M
Type of Event	3
Casualty	
Rainfall (mm)	

Slide ID	Newspaper&Media/2020/Sikkim/136
Source	Newspaper&Media
Slide No	136
State	Sikkim
District	South Sikkim
Name	Jorengthang Landslide
Location	Road connecting DANs Energy Power Station to NH Jorethang Namchi, IPCA Lab damaged
NH-SH Cutoff/Affected	NH510
Lat (dd)	27.163
Long (dd)	88.311
Date of Event	6-Aug-20
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	31.63062681

Slide ID	Newspaper&Media/2020/Sikkim/137
Source	Newspaper&Media
Slide No	137
State	Sikkim
District	East Sikkim
Name	East_sikkim_Is_Sept_2020
Location	Sirwani Sliding area, Tirteray Dhara, Lower Sudungkhala
NH-SH Cutoff/Affected	NH10
Lat (dd)	27.198
Long (dd)	88.648
Date of Event	8 to 11 September 2020
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	99.18681894

Slide ID	Newspaper&Media/2020/Sikkim/138
Source	Newspaper&Media
Slide No	138
State	Sikkim
District	East Sikkim
Name	North_Sikkim_Is_Sept2020
Location	Mangan to Gangtok State Highway via Rakdung-Tintek Road
NH-SH Cutoff/Affected	NH10
Lat (dd)	27.393
Long (dd)	88.515
Date of Event	8 to 11 September 2020
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	12.22244298

Slide ID	Newspaper&Media/2020/Sikkim/139
Source	Newspaper&Media
Slide No	139
State	Sikkim
District	South Sikkim
Name	South_Sikkim_Is_Sept2020_1
Location	Namchi to Jorethang Road at 12th mile, Jorethang to Melli road is blocked at Charcharey, near Rolu
NH-SH Cutoff/Affected	NH510
Lat (dd)	27.116
Long (dd)	88.323
Date of Event	8 to 11 September 2020
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	60.38755579

Slide ID	Newspaper&Media/2020/Sikkim/140
Source	Newspaper&Media
Slide No	140
State	Sikkim
District	South Sikkim
Name	South_Sikkim_Is_Sept2020_2
Location	The Kewzing to Shakbari road is also blocked at Tingmoo
NH-SH Cutoff/Affected	NH710
Lat (dd)	27.254
Long (dd)	88.317
Date of Event	8 to 11 September 2020
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	45.35731913

Slide ID	Newspaper&Media/2020/Sikkim/141
Source	Newspaper&Media
Slide No	141
State	Sikkim
District	South Sikkim
Name	West_Sikkim_Is_Sept2020
Location	Reshi to Nayabazar road being blocked at Kaijaley and Nagdhara
NH-SH Cutoff/Affected	NH510
Lat (dd)	27.082
Long (dd)	88.429
Date of Event	8 to 11 September 2020
Time of Event	
Type of Event	3
Casualty	
Rainfall (mm)	108.4387225

ANNEXURE-II: NEWSPAPER CLIPPINGS

ARUNACHAL PRADESH

The Shillong Times

ESTABLISHED 1945

Home > News Alert >

4 of a family die in landslide in Arunachal

By From Our Correspondent... On Jul 10, 2020

NEWS ALERT REGIONAL



Representative picture

GUWAHATI: Four members of a family were killed in a landslide in Arunachal Pradesh.

The incident took place at Tigdo village in Arunachal's Papumpare district on Friday morning.

Chief Minister Pema Khandu condoled the death of the four family members who died in the landslide.

'May the deceased rest in eternal peace. My sincere condolences to the bereaved families,' Khandu tweeted.

The Shillong Times

ESTABLISHED 1945

Home > News Alert >

Minor girl killed in landslide, lady drowned in Itanagar

By From Our Correspondent... On Jun 25, 2020

NEWS ALERT REGIONAL



Representative picture.

GUWAHATI: A minor girl was killed after getting buried in a landslide that occurred at Doni Colony in Itanagar in Arunachal Pradesh on Thursday morning while a lady was drowned in Lagun River near Naharlagun on Wednesday evening.

Arunachal woman, her two kids buried alive in landslide

The sources said incessant rains for the past two days that triggered the landslides and also flood left trails of large-scale devastation in the entire state.



Published: 26th May 2020 06:33 PM | Last Updated: 26th May 2020 06:33 PM

🔊 | A+ A A-



For representational purposes

By Prasanta Mazumdar

Express News Service

GUWAHATI: Three of a family died in a landslide, triggered by incessant rains, in Dibang Valley district of Arunachal Pradesh.

Official sources said the incident occurred at Arzoo village at around 2 pm on Monday. A 27-year-old woman and her son (7 years) and daughter (3 years) were buried alive when their house came under the landslide.

The sources said incessant rains for the past two days that triggered the landslides and also flood left trails of large-scale devastation in the entire state.

Chief Minister Pema Khandu mourned the deaths. He announced ex-gratia of Rs 4 lakh each to the next of kin of the victims.

Arunachal: Landslide on NH-15 buries 5 trucks at Banderdewa

AN by NE NOW NEWS , September 18, 2020 6:14 pm



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Five trucks were buried while a shop was damaged on Friday in a massive landslide triggered by heavy rains on NH-15 at Banderdewa along Assam-Arunachal Pradesh border.



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PUBLISHED SIMULTANEOUSLY FROM GUWAHATI & DIBRUGARH

RN-112/57 TECH/GH - 103/2018-2020, VOL. 82, NO. 149

www.assamtribune.com

Pages 12 Price: ₹ 6.00

GUWAHATI, WEDNESDAY, JUNE 3, 2020

p2 EU, UK leaders to meet as post-Brexit talks stall

p5 CM reviews GMC's solid waste management system

p11 Govt to initiate more reforms to get growth back

NEWS Capsule

Modi-Trump discussion

NEW DELHI, June 2: Prime Minister Narendra Modi spoke with US President Donald Trump today and discussed the COVID-19 pandemic, among other issues. ... We discussed his plans for the US Presidency of G-7, the COVID-19 pandemic, and many other issues," Modi tweeted. — PTI

Chargesheet

NEW DELHI, June 2: Delhi Police crime branch today filed a chargesheet against Pinarji Todi activists Natasha Narwal and Devangana Kalita in connection with the Jafabad riot case. The court has granted bail to Devangana Kalita. — Spt Correspondent

World outrage

BRUSSELS, June 2: From Sydney to Paris, world outrage at George Floyd's death in the United States was growing on Tuesday as the European Union's top diplomat said the bloc was "shocked and appalled" by it and thousands marched in Australia's Sydney. — AP

Appointment

NEW DELHI, June 2: Seasoned diplomat Gauri I Kumar has been appointed as India's next High Commissioner to the United Kingdom, the External Affairs Ministry said on Tuesday. Kumar, a 1986-batch Indian Foreign Service officer, succeeds Ruchi Ghanshyam. — PTI

2 ultras shot

Landslide claims 21 lives in Barak valley



District administration and police officials with local people at a landslide site in Karimganj district, on Tuesday. — UB Photos

STAFF CORRESPONDENT

SILCHAR, June 2: Twenty one people were killed in separate incidents of landslide triggered by heavy rain in the Barak valley on Tuesday. The deceased include seven from Cachar district, eight from Hailakandi and six from Karimganj. Among the dead eight are minors, sources said.

Cachar deputy commissioner Keerthi Jali told *The Assam Tribune* that rescue operations are going on, while arrangements are being made to provide ex gratia to the kin of the deceased. Shamim Ahmed Laskar, project officer of district disaster management authority, said the Cachar incident took place in the Joyrupa area. "All the de-

Karimganj deputy commissioner Anamulhan MP informed that six persons, including four children, from two different families in the Karimganj area under Kaligunj police outpost lost their lives due to landslide.

"While all five members, including three minors, of one family got trapped inside their *kutcha* house due to the landslide, another death of a 14-year-old girl occurred around 2 km away in another landslide," the official said.

In the second landslide incident, one member of the family sustained leg injury and had been moved to hospital for urgent medical attention. Requisite steps are being taken to provide succour to the affected families, the deputy commissioner said.

morning. Of the eight dead, five belonged to the same family.

Landslides occurred in the three different places of Aigapur and Mohanpur village of the district around 8 am. Local people rescued the injured before the SDRF team could reach the spots.

Meanwhile, the State government has announced payment of Rs 4 lakh each to the next of kin of the deceased. Staff Reporter adds from Guwahati: Assam Governor Jagdish Mukhi and Chief Minister Sarbananda Sonowal today expressed grief over the loss of lives in the incidents of landslide in the three districts of Barak valley.

Sonowal directed the deputy commissioners of the districts to speed up rescue operations and extend help to the

Chinese troops moved into eastern Ladakh, says Rajnath

NEW DELHI, June 2: A "sizeable number" of Chinese troops have moved into eastern Ladakh and India too has taken all necessary steps to deal with the situation, Defence Minister Rajnath Singh said on Tuesday, referring to the nearly month-long high-altitude standoff between the two armies.

Singh said a meeting between senior Indian and Chinese military leaders has been scheduled for June 6 even as he asserted that India is not going to back off from its position.

Asked about the current situation in sensitive areas in eastern Ladakh, he said the Chinese have come up to what they claim is their territory while the Indians believed it is theirs.

"There has been a disagreement over it. A sizeable number of Chinese people have come there. India has done what it needs to do," Singh told CNN-News 18. Singh's comments were seen as first official confirmation of presence of significant number of Chinese troops in the disputed areas which India maintains are on its side of the Line of Actual Control (LAC), the de-facto border between the two countries.

According to reports, significant numbers of Chinese troops have been coming in the Indian side of the LAC in Galwan Valley and Pangong Tso. The Defence Minister said China should think about the issue seriously so that it can be resolved soon.

Indian and Chinese troops were engaged in a bitter standoff in several areas along the Line of Actual Control in mountainous eastern Ladakh for close to a month. Both the countries are holding talks at military and diplomatic levels to resolve the dispute.

"The Doklam dispute was resolved through diplomatic and military talks. We have found solutions to similar situations in the past as well. Talks at the military and diplomatic levels were on to resolve the current issue," Singh said.

"India does not hurt pride of any country and at the same time, it does not tolerate any attempt to hurt its own pride," he said, talking about India's long-held policy.

SEE PAGE 6

State flood situation improves

STAFF REPORTER

GUWAHATI, June 2: Even though the flood situation in the State showed further signs of improvement on Tuesday, around 1,55,850 people in three districts — Goalpara, Nagaon, Hojai — are still reeling under waters.

The daily flood report of the Assam State Disaster Management Authority (ASDMA) said Goalpara is the worst hit with 88 villages/localities under water affecting 1,18,548 people. In Nagaon, 8,670 people were affected in 93 villages and in Hojai 28,634 people were affected in 58 villages.

Among the flood affected, a total of 6,122 people are taking shelter in 40 relief camps across the districts.

Floodwaters have submerged 11,987 hectares of crop land in Goalpara, 5,754 hectares in Nagaon and 6,572 hectares in Hojai district. In past 24 hours, flood damaged an embankment and a number of roads in Karimganj district. Incidents of road damage have also been reported from Barpeta district.

Moreover, flood-triggered erosion was reported from Dibrugarh and Biswanth districts. Incidents of urban flood were also reported in Cachar district.

COVID 19

The Shillong Times

ESTABLISHED 1945

Home > News Alert >

2 youth buried alive in landslide at Jorabat

By Special Correspondent

On Jul 12, 2020

NEWS ALERT REGIONAL



GUWAHATI: Two youths were buried alive by a landslide triggered by heavy overnight rainfall in the 12th Mile area of Jorabat early Sunday morning, official sources informed.

Landslide on NH-37 Manipur: disturbs traffic for some hours

By UT News Service August 28, 2020 - 9:32pm



IMPHAL: Normal vehicular movement along the Imphal-Jiribam road (NH-37) was disrupted for a few hours on Friday due to a massive landslide at Kotlen.

Reports said that the road was totally blocked by sliding rocks around 11 am, forcing all kinds of vehicles moving on the highway to stop on both sides.

The rocks and mud were, however, cleared by a team of National Highways & Infrastructure Development Corp Ltd (NHIDCL) who swung into action immediately. The team was assisted by police personnel of New Keithelmanbi police station.

With the clearance of the debris that blocked the road, normal traffic resumed around 4 in the evening, the report said.

The debris was cleared only to allow the stranded vehicles and works to completely clear the debris was going on till late in the evening, the report also said.

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Press Trust of India
Imphal
May 27, 2020 UPDATED: May 27, 2020 14:14 IST

Image for Representation

Massive landslides triggered by heavy rain blocked the Imphal-Jiribam stretch of the national highway (NH) 37 in Manipur on Wednesday, police said.

Heavy rain lashed Manipur on Tuesday night. Landslides blocked the NH 37 between Nunga and Rengpang at around 3 am, they said.

At least 10 buses carrying stranded Manipuris who arrived at Jiribam railway station in Shramik special train were stranded due to the landslides, a senior police officer said.

The police personnel of Nungba Police Station along with local volunteers and drivers cleared the debris after which the buses could proceed towards Imphal, said the officer.

The heavy rain also damaged several temporary constructed structures including the scanning booth at Jiribam Railway Station, officials said.

Jiribam is the only railway station in Manipur and all the shramik special trains carrying thousands of stranded Manipuris are arriving at the station. From Jiribam the people arriving in train are sent to their respective districts in buses. **LIVE TV**

THIS STORY IS FROM OCTOBER 26, 2020

Manipur: Landslide in Senapati cuts off connectivity via National Highway-2

ANI | Updated: Oct 26, 2020, 08:37 IST



A landslide that occurred on October 23 in Senapati district has cut off connectivity via national highway-2 (... Read More)

SENAPATI: A landslide that occurred on October 23 in Senapati district has cut off connectivity via the National Highway-2, said an official on Sunday.

Jayson, project manager, BIPL said that the administration was trying its best to resume the connectivity but it may take longer if the weather remains unfavourable.

Landslide causes extensive damage to Khenjonglang Theatre Complex

Source: The Sangai Express

Imphal, August 11 2020: Persistent moderate rainfall for the last one week in various parts of the State have resulted in landslide of the Awa Ching located at Keirao Khunao of Imphal East district today, causing extensive damage to the Khenjonglang Theatre Complex located adjacent to the hillock.



The complex suffered damage when mud, boulders and other debris carried by the landslide hit the theatre complex around 4:00 am.

Dr S Thaninleima, Director, Khenjonglang Theatre said that the 89 sq ft wide stage has been damaged to a great extent by the landslide and it is unlikely for the stage to be usable without repairs.

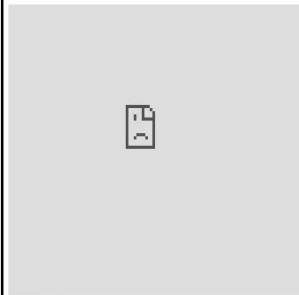
The dining hall, dressing hall, plus musical instruments, lights and other theatre props have also suffered damages to the tune of Rs 11 Lacs, she added.

The Director further added that details about the accident will be reported to the Iribung Police Station as soon as possible and urged concern authority to look into the matter.

Landslide halts vehicular movement

Source: The Sangai Express

Moreh, August 20 2020: Owing to landslides that were reported from certain areas of Khongkhang along Imphal Moreh road and Tengnoupal district, vehicular movements along the Highway stretch was halted for a brief period, this morning.



Landslide halts vehicular movement

It is believed that the landslides might be caused by the incessant rainfall that is witnessed in the State since the last two-three days.

The section of highway stretch was opened for vehicular movement after the earth/debris was cleaned using JCBs of GR Infra Project Company Limited, which is engaged for Imphal Moreh road widening project.

Meghalaya's Shillong bypass closed due to landslide

by NE NOW NEWS September 27, 2020 8:37 pm



The landslide occurred on Saturday due to the collapse of big stones from a stone quarry near the bypass road.

Share on Facebook

Share on Twitter

The national highway via Shillong bypass has been closed for vehicular traffic following massive landslide at Thangshalai in Meghalaya's East Khasi Hills district.

Officials in the National Highway Authority of India (NHAI) on Sunday said that the landslide occurred on Saturday due to the collapse of big stones from a stone quarry near the bypass road.

Roads cut off in SWKH; flash flood, landslide in Nongstoin

By From Our Correspondent On Jul 12, 2020

MEGHALAYA



An excavator at work during the repair of the wing wall at Mawkyrwat-Marshillong road on Saturday.

Mawkyrwat/NONGSTOIN: A footbridge was washed away and roads were cut off while reports of landslides occurred in various parts of South West Khasi Hills due to incessant rainfall in the district for the past few days.

In Pictures

IN PICTURES | NH-06 Sohbhala badly affected by landslides

NH-06 Sohbhala, Umsning in Ri-Bhoi district of Meghalaya badly affected by landslides.



The Shillong Times

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2 killed in landslide in Shillong

By By Our Reporter On Sep 24, 2020

MEGHALAYA NEWS ALERT



SHILLONG: Two people were killed in a massive landslide in Police Reserve in the city on Thursday.

Massive landslide in Meghalaya

by NE NOW NEWS SHILLONG, July 11, 2020 10:21 am



Image courtesy: Twitter @MeghalayaPolice

Share on Facebook

Share on Twitter

Meghalaya has been experiencing heavy landslide in between Laitlyngkot and Pomlum along National Highway 40 since Friday.

MOKOKCHUNG

Landslides destroy houses and roads in Mokokchung

By EMN
Updated: Aug 16, 2020 7:00 pm

A | A | A



A house partially affected by landslide at Alongmen ward in Mokokchung town on August 15.

Mokokchung, Aug. 16 (EMN): Incessant torrential monsoon rains in Mokokchung have triggered landslides in and around the district and also affecting the lives of the people.

A press statement on Sunday informed that landslides had completely washed away a connecting road in Alongmen ward in Mokokchung town on August 15 night. Besides, it has affected more than thirty households of the particular ward.

Upon hearing about the incident, the minister of Housing and Mechanical Tongpang Ozukum visited the impacted area on August 16 morning in order to take stock of the situation. He has assured to provide assistance from the state government in order to restore normalcy at the earliest.

The minister was accompanied by the deputy commissioner, Mokokchung, Limawabang Jamir, superintendent of police, Mokokchung, Vishal; SDO (C) Tamsuchuba; officials from NSDMA and a host of other officials, it stated.

Incessant rain damages houses in Kohima



NAGALAND | 20th August 2020

Our Correspondent
Kohima | August 20

Incessant rain for the past few days has caused landslide affecting atleast three private houses in Merhülietsa, Paramedical colony in Kohima.

Massive landslide occurred on Wednesday at around 11:45 am bringing down portion on land with plantation blocking National Highway-29 near Timber Mill.

John Sekhose said that incessant rain caused landslide damaging portion of his timber house and also affecting a RCC house on NH-29.

He said following the expansion work of NH-29 and construction of retaining wall, the area has been witnessing landslides every monsoon since 2018.

He recalled that in 2018 one of his timber houses was totally damaged, while Wednesday's landslide has put another timber house on the verge of collapse and also affected an Assam type house.

ZUNHEBOTO

Landslide damages several houses in Zunheboto

By EMN
Updated: Aug 17, 2020 8:00 pm

A | A | A



Seen in the picture are houses damaged by a landslide at Zunheboto town.

Dimapur, Aug. 17 (EMN): At least 5 houses were damaged by landslide triggered by heavy rains in the morning of August 17, at DC Hill West Colony in Zunheboto town.

DIMAPUR

Massive landslide near Chümoukedima check gate affects traffic, one car damaged

By Our Reporter
Updated: Sep 19, 2020 6:12 pm

A | A | A



Our Reporter

Dimapur, Sep. 19 (EMN): A massive landslide occurred near Chümoukedima check gate in Dimapur on Saturday morning, a traffic official informed.

The official said that the landslide occurred about three kilometres after the check gate, at around 5 am.



The traffic official informed that one-way has been cleared for vehicular movement, during the filing of this report.

Incessant rain cause landslide, damage properties in Kohima



Landslide damaged around 10 houses on Sunday night. (NP)

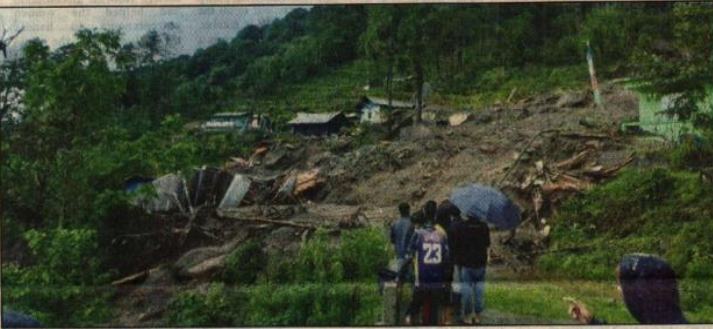
Correspondent KOHIMA, OCT 5 (NPN) | Publish Date: 10/5/2020 1:14:05 PM IST

Incessant rain for the past few days in Kohima caused landslide and damaged 10 houses at Lower PWD at around 9:30 pm on Sunday.

Speaking to Nagaland Post on Monday, Lower PWD Youth Organization president, Kevi Mere informed that around 30 youths volunteered to help the aggrieved families affected by the landslide. He said that the area was prone to landslide but the major disaster occurred last night damaging properties.

He also informed that the incident was reported to the deputy commissioner Kohima and the District Disaster Management Authority, however no official has visited the site to take stock of the situation.

Mudslides claim one life, two injured in North



SE Report

GANGTOK, May 24: One person was killed and two others injured in a landslide triggered by continuous rainfall at Upper Lingchom under Kabi-Lungchuk constituency today.

As per the information received by IPR Mangan from the Kabi SDM, the landslide occurred at around 1 pm killing a 50-year-old woman while trying to escape the under Kabi-Lungchuk constituency today.

The deceased has been identified as Phu Phu Sherpa, a resident of Upper Lingchom, North Sikkim. Lakpa Doma Sherpa and Bhairung Sherpa suffered injuries. They have been taken to STNM Hospital, Gangtok for treatment.

Further, the landslide damaged three houses. Kabi SDM Prem Gurung and officials immediately reached to the spot and evacuated other people and shifted them to the safer place, the IPR release informs.



Two labourers buried alive

NAMO.DIXIT

NAMCHI, June 8: Two labourers were killed when mud caved in on them while they were working on a road widening and wall construction projects at Thaka Road, Ben, South Sikkim. The incident took place on Saturday afternoon when

around 13 labourers were engaged by the road contractor for widening of the road and wall construction. Many escaped from the mudslide with injuries while two of them died on the spot. The deceased have been identified as Dhan Bahadur Tamang (40) of Birpara, West Bengal and Dil Bahadur Chettri (50) of Lingyong, South Sikkim who were completely buried by the mudslide and boulders resulting in their death. The labourers were engaged in the road construction project by Das and Das Company, a subsidiary of NHIDCL. After the incident, the workers and locals retrieved the dead bodies which were later sent for autopsy.

DZONGU TOTALLY CUT-OFF

Slides fracture road networks, several villages affected

BDOO GURUNG
GANGTOK, June 28: Entire Dzongu is cut-off from rest of the State following multiple slides across the area due to heavy rains for the past few days. The worst-affected area is Passingdong village in Upper Dzongu, about 10 km away from Mangan, the north district headquarters. According to the information available, a massive landslide came down from uphill Panang on Saturday night which damaged around 19 houses, local road and government secondary school at Passingdong. Since the landslide is active and threatens to cause more damage due to continuous rains, the authorities have evacuated the residents of entire village to a safer location. The road from Mangan to Passingdong was also damaged in the morning hours of Saturday. The North Sikkim highway which connects Gangtok with Mangan is also damaged due to slides at some places. Similarly, the Singu-Dikchu-Mangan route is also not reliable at the moment for vehicle movement. Meanwhile, landslides have damaged houses and roads at Lingchi-Barfik CPU and Hoo-Gyathang, both in Upper Dzongu. The entire Dzongu is cut-off due to slides occurring since past couple of days, said the area MLA. Luni in Lower Dzongu also saw damage to houses and land due to landslides. "Due to the extensive landslides and continuous rains, it is not possible to talk about restoration measures at the moment. Right now we can only undertake relief measures since entire Dzongu is affected. Cracks have appeared on lands at many villages that may give way to landslides if the rains do not stop," said the area MLA. Panang is about one km uphill from Passingdong. The hill side gave away following intense rains in the past two days. According to a report from Sikkim investigation division of Central Water Commission based in Gangtok, Sangkhalung - the entry point to Passingdong and other villages of Upper Dzongu - recorded 227.4 mm and 102.4 mm rainfall in the

ENTIRE PASSINGDONG VILLAGE EVACUATED



The landslide left behind a trail of devastation at Passingdong village on Saturday night.



The Passingdong slide had advanced from Panang area, around one km uphill from the village. The slide is still active and poses threat to settlements below.

Area MLA Pato Namgyal Lepcha and district authorities inspecting the damages at the landslide-hit village.

Surviving The Dangerous Saturday Night

BDOO GURUNG
GANGTOK, June 28: As the landslide advanced from uphill Panang at around 8 pm of Saturday night, the 35-odd families of Passingdong village were caught between the devil and deep sea. Heavy rains, complete darkness and a dangerous slide rumbling through their village where could safety be found? Danger lay outside where one could get caught by the slides in the darkness. It was risky and traumatic to stay indoors also considering the intensity of the slide. As shared by the local residents, many of them rushed outside and braving the slides, took shelter in whichever place they could find stable and safe. Others decided to stay where they were given it was pitch dark outside amidst a ferocious landslide. "We tried to evacuate in the night but there was no power, it was pitch dark and continuous rains at that moment. Some of the villagers managed to evacuate and took shelter elsewhere while others decided to stay where they are," said local village Gyatso Lepcha. In the morning hours of Sunday, the images of a devastated Passingdong village started circulating in social media depicting the intensity of the landslide. The district authorities reached the village and despite logistic issue due to fractured road connectivity, have managed to set up a shelter camp at the nearby PHC. The whole village has been evacuated and nobody is staying in the village at the moment, said Gyatso Lepcha. The danger is still there, one can see huge boulders are still hanging precariously on top of the slide uphill which can come down anytime, said Gyatso. We have to wait for the slide to be stabilised before any restoration work can be taken up, he added. A similar landslide had damaged houses at Passingdong in 2018. The slide this time was three times in intensity, said the local resident. Gyatso informed that even the Miner & Geology report shows that Passingdong lies in a vulnerable area and prone to landslides. We may later approach the government requesting that the people of Passingdong are resettled at another safer location, he added. Further above Passingdong lies Manton which had been completely devastated by a massive hill slide in August 2016. Since then, there is no permanent road connectivity to Tingyong and Salyang-Pemung CPU.

Landslides cut off North Sikkim

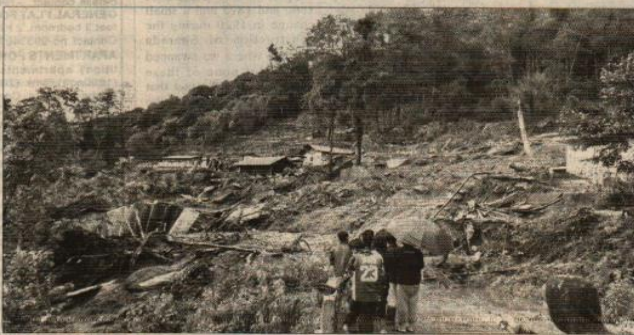
SUMMIT REPORT

GANGTOK, 25 JUN: North Sikkim was cut off from rest of the State Thursday morning following incessant rains triggering landslides at several places along the North Sikkim Highway.

North District Magistrate, Tenzing T. Kaleon informs that North Sikkim Highway was blocked at several places at Ambithang, near Rangrang and Ramthang under Namok Gram Panchayat Unit, Phamtham at Kabi Lungtsok connecting North Sikkim from Tashi View Point and Dikchu-Mangan road, at Tintek and Dikchu Sokpay.

He added that no casualties or damage to property have been reported so far. Mr Kaleon informed that it will take a few days to fully restore road connectivity and the District Administration has directed all the officials to be on alert.

One died, two injured in Lingchom landslide



Gangtok, May 25: The incessant rain for the last couple of days brought havoc to North Sikkim especially some parts of Kabi-Lungtsok constituency where a lady has been killed by a landslide that

occurred right above the village at Lingchom ward at around 12:30 pm on May 25. The deceased is identified as Phu Phu Lhamu Sherpa, age 57 year and the wife of Samdup Sherpa of Upper Lingchom.

The deceased and her family members had gone for shelter to their neighbour's house when the first landslide triggered and washed away the toilet of her home this morning. The second landslide that

occurred right above the house of Passang Sherpa where the deceased and the family were sheltered was also damaged that killed Phu Phu Lhamu Sherpa and injured two others. The body has been sent for autopsy at District Hospital Mangan. The DC North, TT Kaleon, area MLA cum Minister Forest, Karma Loday Bhutia, SDM Kabi along with BDO and SHO Phodong Police Station visited the spot to take the stock of damages. Meanwhile, the two injured persons have been rushed to STNM Hospital Gangtok. The landslide also blocked Gangtok-Mangan road at Phensong and near Kabi today. The approach road leading to Upper Lingchom about 4 km from main road of Mangan-Gangtok highway has been inaccessible due to the landslide at several areas.

ANNEXURE-III: SDMA REPORT SAMPLE

MIZORAM

Field Landslide Data Collection Format

Details to be filled up by the field officer: Aizawl Landslide (2020) N0 -1 Locality: Zemabawk

Sl No.	Details		Attribute / Remarks
1	History of landslide	a	New
		b	Old✓
2*	Date of Event		6-10-2020
3*	Time of Event		3:05 pm
4	District Name		Aizawl
5	State Name		Mizoram
6	Length (m)		60m
7	Width		40m
8	Height		35m
9*	Latitude/Longitude		23°43'54.94" N 92°45'36.10" E
10	Type of Landslide	a	Debris Slide✓
		b	Debris Flow✓
		c	Rock fall
		d	Rock slide
		e	Others
11	Material involved	a	Rock – Crown is located on rocks
		b	Overburden – Crown is located on weather material or soil✓
		c	Both – both of the above
12	Occurrence	a	Road cutting✓
		b	River ban
		c	Natural hill slope✓
		d	Opposite valley
		e	Others
13	Structure affected	a	Road or railway line✓
		b	Houses✓
		c	Land (barren or cultivated)
		d	Forest✓
		e	Civil project
		f	Others
14	Casualties	a	Human
		b	Livestock
		c	Both
15	Number of casualties		
16	Trigger of landslide	a	Rainfall✓
		b	Earthquake
		c	Others✓
17	Rainfall	a	No rainfall
		b	Cloudburst
		c	24 hrs✓
		d	One week
		e	Other✓ Rainfall during the day and whole week before
18*	Amount of rainfall (mm)		54.5 mm
19	Cause of landslide		Heavy rainfall and Improper drainage, slope cutting for construction of road
20	Remedial measure		Drainage control and breast wall

LANDSLIDES IN 2020							
Sl.No	Name of the slide	Lat	Long	Toposheet	Locality	Date of Incidence	Casualty
1	Chandipur Grant Slide	24.79488889	92.55505556	83D/9	Chandipur, Bowarthar, Algapur Circle, Hailakandi	2/6/2020	1 dead
2	Mohanpur Grant Slide 3	24.73116667	92.64738889	83D/10	Mohanpur Grant, Hailakandi Circle, Hailakandi	2/6/2020	1 dead
3	Mohanpur Grant Slide 2	24.70655556	92.65008333	83D/10	Mohanpur Grant, Hailakandi Circle, Hailakandi	2/6/2020	1 dead
4	Kanakpur Slide	24.92097222	93.07305556	83H/1	Kanakpur Pt III, Lakhipur, Cachar	2/6/2020	7 dead
5	Govindanagar Slide	24.72191667	92.90380556	83D/14	Govindanagar Pt. II, Sonai, Cachar	5/6/2020	Nil
6	Mohanpur Grant Slide 1	24.70363889	92.64680556	83D/10	Mohanpur Grant, Hailakandi Circle, Hailakandi	2/6/2020	5 dead
7	Karimpur Slide 1	24.78380556	92.45563889	83D/5	Karimpur, Nilam Bazar, Karimganj	2/6/2020	1 dead
8	Karimpur Slide 2	24.78002778	92.45372222	83D/5	Karimpur, Nilam Bazar, Karimganj	2/6/2020	5 dead
9	Kahilipara Slide	26.146	91.756	78N/16	Power grid, Kahilipara, Guwahati	6/26/2020	1 dead
10	Amingaon Slide 01	26.21083333	91.68944444	78N/12	Amingaon, Guwahati	6/25/2020	Nil
LANDSLIDES IN 2019							
Sl.No	Name of the slide	Lat	Long	Toposheet	Locality	Date	Casualty
1	Kashmipur Slide	25.32069444	93.12733333	83G/03	Kashmipur, Maibong, Dima Hasao District	8/12/2019	Nil
2	Digarur slide	26.14538889	91.98458333	78N/16	Indian Air Force Station, Digarur, Kamrup District, Assam	9/29/2019	Nil
3	Amiya Nagar Slide	26.19236111	91.78080556	78N/16	Amiya Nagar, Chandmari, Guwahati	7/10/2019	Nil
4	Bezbaruah Nagar slide	26.15805556	91.71258333	78N/12	Nilachalpur, Bezbaruah Nagar, Guwahati	7/16/2019	Nil
5	Katahbari slide	26.11816667	91.71330556	78N/12	Katahbari, Garchuk, Near Bethany High school, Guwahati	7/10/2019	1 dead
6	Matia pahar slide	26.10591667	91.62283333	78N/12	Natun Basti, Matia Pahar, Dipar Bil	7/10/2019	1 dead
					Note: Casualty mentioning only dead.		
SL NO	DATE	TIME	DISTRICT	REVENUE CIRC	VILLAGE NAME	COORDINATES	
1	6/8/2020		CACHAR	SONAI	Panichowki & Saidpur 4	24 38 52.35 92 59 18.07 , 24 38 49.14 92 59 20.58, 24 38 59.60 92 59 20.67, 24 43 40.36 92 49 23.43	
2	6/26/2020	4:00 PM	KAMRUP RURAL	NORTH GUWAHATI	Gauripur	26.214209 91.693519	
3	6/27/2020	6.00 PM	KAMRUP RURAL	NORTH GUWAHATI	DIRGHESWARI	26.254368 91.762678 26.262201 91.759778	
4	6/4/2020		CACHAR	SONAI	GOBINDANAGAR PT II	24 43 18.74 92 54 13.89	

ANNEXURE-IV: GSI REPORT SAMPLE

NOTE ON THE POST-DISASTER PRELIMINARY ASSESSMENT OF LANDSLIDE AT TIGDO VILLAGE, PAPUM PARE DISTRICT, ARUNACHAL PRADESH

By
Hiruj Saikia, Senior Geologist
Gangesh Das, Assistant Geologist

INTRODUCTION:

In compliance of official order received from Dy. DG SU: AP, landslide assessment around Papum Pare district, Arunachal Pradesh in parts of T.S No 83E/12 were carried out on 17.07.2020. As per the media reports, landslide triggered due to heavy rain occurred at Tigdo Village under Doimukh circle, at around 2:30 am on 10.07.2020. As a result of landslide, a house was totally damaged and 4 persons of a family buried and causes loss of property. Preliminary assessment was carried out to evaluate the causative factors and to suggest site specific remedial measures. During the field assessment relevant information pertaining to the date and time of occurrence and destruction of property and loss of lives were collected from the villagers and district administration.

Post-disaster landslide data was recorded using 42-point geo-parametric datasheet. Information from locals was also collected.

GEOLOGY:

The area lies in the Lower Himalayan Belt and exposes the rocks of Kimin Formation of Siwalik Supergroup and recent deposits of unoxidized sand and pebbly horizons of Quaternary. The Kimin Formation consists of grey, soft, medium to coarse grained sandstone, often with pebbly and gritty horizons. The sandstone consists of primary structures such as cross bedding and is poorly graded from coarse to fine grained and associated with pebbles.

CAUSATIVE FACTORS:

The slide is of translational type having length, width and height of 7m, 16m and 10 m respectively with a run out distance of 22 m. The slide material is mainly of unconsolidated sand, silt, clay and pebbles.

The continuous heavy rain of monsoon has triggered the landslide. Subsurface percolation of rainwater resulted the build-up of undesirable pore pressure in overburden material causes the reduction of the shear strength of the slope forming material. The main causative factors of this landslide are also attributed to unconsolidated sand, silt and pebble horizons, anthropogenic activities, such as deforestation and high angle slope cutting. The slope of the landslide facing is S65°W. As observed, the slope cutting is about 65° – 70°, which is high

ANNEXURE-V: INVENTORY ATTRIBUTES

Slide ID	Unique Identifier for Landslide
Source	Newspaper/Media, SDMA, GSI or From Field
Slide No	Serial number
State	State in NER where landslide occurred
District	District where landslide occurred
Name	Name of the event
Location	Specific location from SDMA reports. General location – neighbourhood, village name etc from newspapers
NH/SH Cutoff/Affected	If National Highway, State Highway or any other important road was cutoff or affected
Latitude	Exact Latitude from GSI/SDMA reports General Latitude from Newspaper Articles
Longitude	Exact Longitude from GSI/SDMA reports General Longitude from Newspaper Articles
Date of Event	Date of the event
Time of Event	Time of event wherever possible
Type of Event	Type 1, 2 or 3 (explained in the next section)
Casualty	Casualty figures
Rainfall (24 hrs)	24 hours rainfall prior to the event

ANNEXURE-VI: CODE SAMPLE

```
*C:\Users\NESAC\Desktop\2020\python\gee_code.py - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

gee_code.py
1 import ee
2 from ee_plugin import Map
3
4 #Code for displaying Pre and Post Imagery
5
6 # Processing polygon_ls
7 polygon_ls = "users/tusharmontfort/senapati"
8
9 aoi = ee.FeatureCollection(polygon_ls);
10
11 # ENTER THE TIME FRAME for the first image
12 start_date1 = '2020-10-01'
13 end_date1 = '2020-10-23'
14
15
16
17 # Import the Sentinel 2 collection as sentinel 2
18 sentinel2_1 = ee.ImageCollection("COPERNICUS/S2")
19
20 # Import the boundary
21 ls_bnd = ee.FeatureCollection(polygon_ls)
22
23 # Filter the image collection using filterBounds() and filterDate() method.
24 # Sort the collection by cloud cover metadata
25 image1 = sentinel2_1 \
26     .filterBounds(ls_bnd) \
27     .filterDate(start_date1, end_date1) \
28     .sort('CLOUDY_PIXEL_PERCENTAGE', False) \
29     .map(lambda image1: image1.addBands(image1.metadata('system:time_start')))\
30     .mosaic()\
31     .clip(ls_bnd)
32
33
34 #TRUE COLOUR COMPOSITE
35 visualization_TCC = {
36     'min': 0.0,
37     'max': 3000,
38     'bands': ['B4', 'B3', 'B2']
39 };
40
41 # FALSE COLOUR COMPOSITE
42 visualization_FCC = {
43     'min': 0.0,
44     'max': 3000,
45     'bands': ['B8', 'B4', 'B3']
46 };
47
48 LS_Scene = ee.Image(image1).select(['B4', 'B3', 'B2', 'B8']);
49
50 Map.addLayer(LS_Scene, visualization_TCC, 'true_color_composite_pre');
51 Map.addLayer(LS_Scene, visualization_FCC, 'False_color_composite_pre');
```